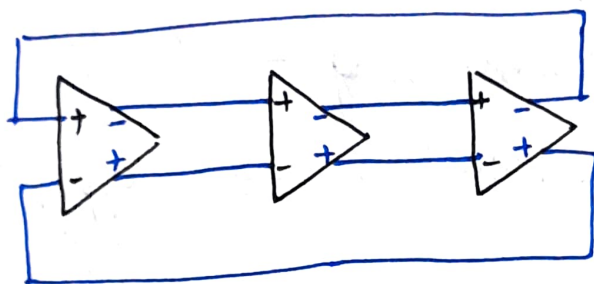


Design of Differential & Multiphase RO

- * 3rd chap ring osc suffer from supply rejection noise
- * differential rings : less sensitive to supply noise
- * Same flow as chapter-3. (1st chap needed to review)

General Consideration.

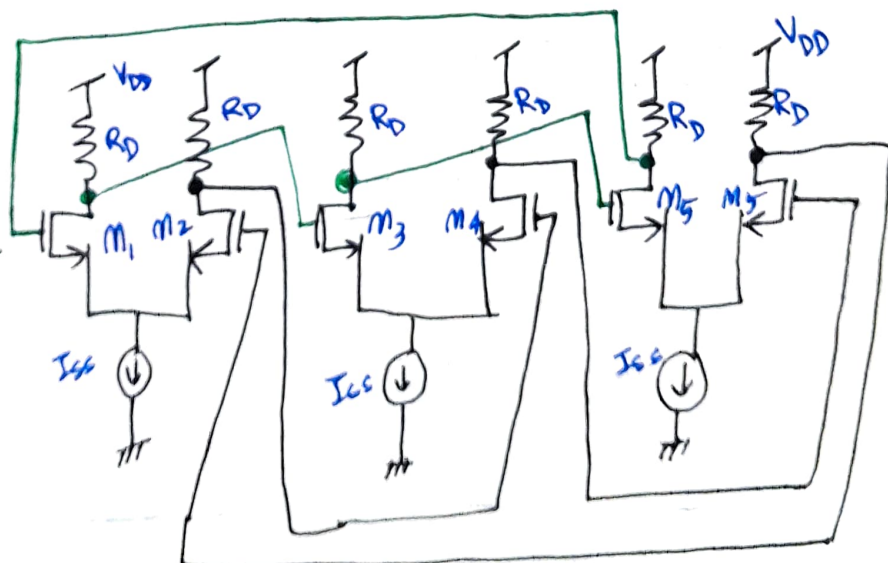


→ Each stage : 60° phase shift

→ low voltage freq gain = 2 (Each stage)

- The circuit produces 6 output phases in 60° steps.

→ -ve feedback at low freq prevents latch-up.



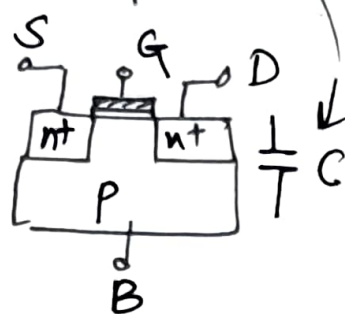
Why differential Circuit?

① Supply Noise Rejection:

- Reduce phase noise
- Improves freq stability.
- Reliability $\uparrow$

unlike inverters, small signal resistance at the drain is independent of V_{DD} $\rightarrow$ less sensitivity.

Supply dependence in diff. rings stems on drain-substrate junction capacitance



② Multiphase O/p generation:

Ex: A 3-stage ring $\rightarrow$ 6 o/ps with 60° steps

A 4-stage loop $\rightarrow$ 8 o/p phases with 45° steps/separation

$\rightarrow$ 2 o/ps per stage

v_1^+ & v_1^-

for N-stages:
 $2N$ output nodes

$\rightarrow$ Phase Delay per stage

$$\theta_{\text{per stage}}^\circ = \frac{360^\circ}{2N}$$

$\leftarrow$ loop
 360° p. shift

how are these separate phases used???

- mixers, modulators
- Multi-phase clocks (e.g., pipelining)

In inverters: transistors depend heavily on V_{DD} .

③ higher speed : → Delay/stage is less than CMOS inverter.

Why???

→ (i) Reduced Voltage Swing

- Single-ended voltage swing
- $I_{DSS} R_D$ is less
- not [0 to V_{DD}] anymore

✓ faster transitions

→ (ii) Lower Capacitance load

- Differential stages use resistors NOT PMOS like in CMOS invt.

- Resistors add very little capacitance than PMOS

- charge/discharge faster at Nodes

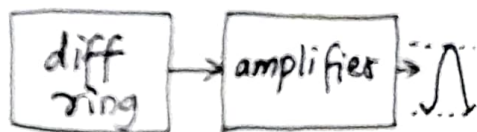
- PMOS add parasitic capacitances → slows down inverters.

Disadvantage ???

Lower Voltage Swing

* Low rail-to-rail voltage swings

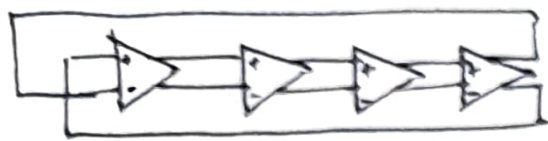
* to achieve full swings



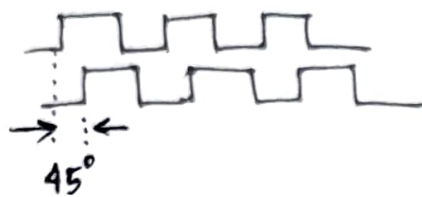
an extra "level-corrector" needed (amp)

✓ increases power consumption

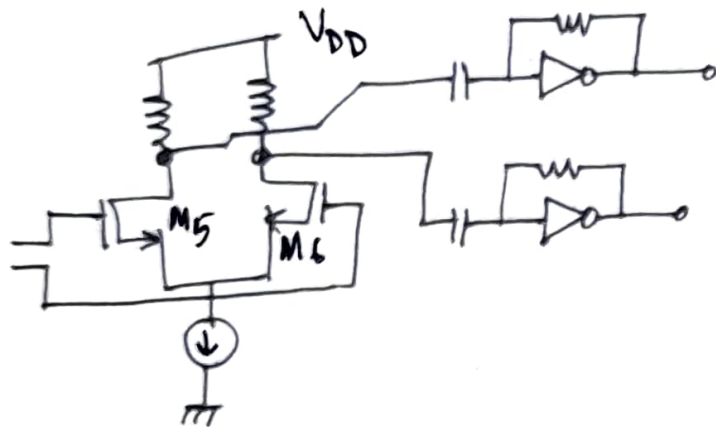
4-stage diff ring $\rightarrow \frac{360}{2 \times 4} = \frac{360}{8} = 45^\circ$



8 output waveforms

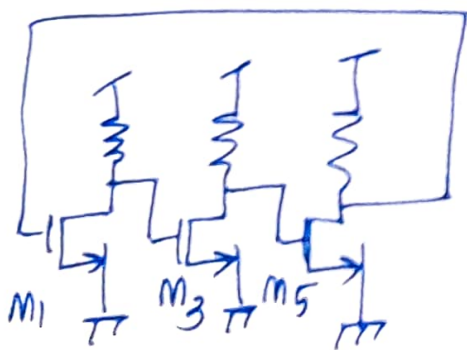


Rail-to-Rail Swings by means of Self-biased
inverters

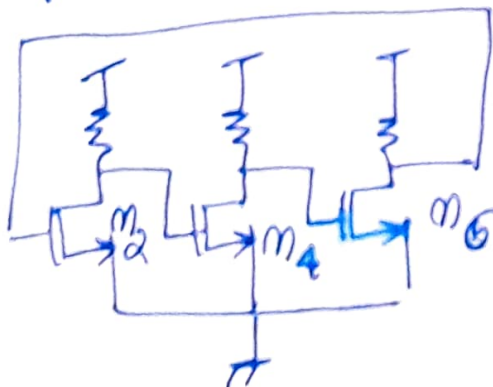


Example 4.1 Role of Tail Current Sources (I_{SS})

* Without the tail current, the circuit splits into 2 single-ended loops independently working their own freq & phases



loop 1



loop 2.

If we replace I_{SS} with a resistor; it will have infinite impedance

✓ if resistance is decreased; (good)

connection becomes poor, two circuit loops split up.

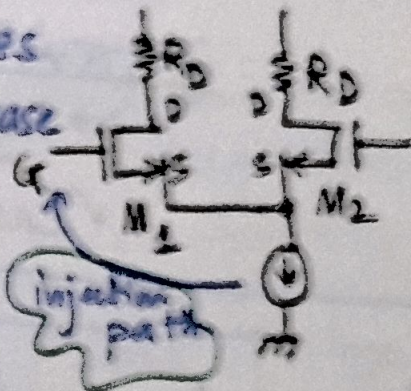
ideal Current Source has infinite impedance
 $Z = \frac{V}{I}$

✓ if resistance is Large, differential op can be maintained as it would mimic ∞ impedance.

Injection Locking - tail current

① where one oscillator influences or locks the frequency & phase of another oscillator.

(here, 2 coupled parts of the same oscillator)



② "injection" happens through common source node.

③ tail current provides "high imp" for injection

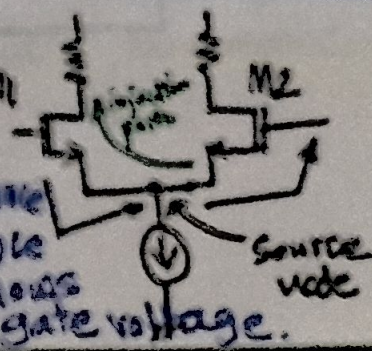
④ of current into each other → to synchronise
operate differently

• M_1 acts as a source-follower

• M_2 acts as a common-gate stage

Input at source
output at its drain
⇒ current gain

high input impedance
low o/p impedance
Source Voltage follows gate voltage.



To behave unified

- injection of current b/w the 2 must be strong to force them into synchronised state
- same freq, 180° phase shift (b/w o/p's).
- A high-impedance tail device

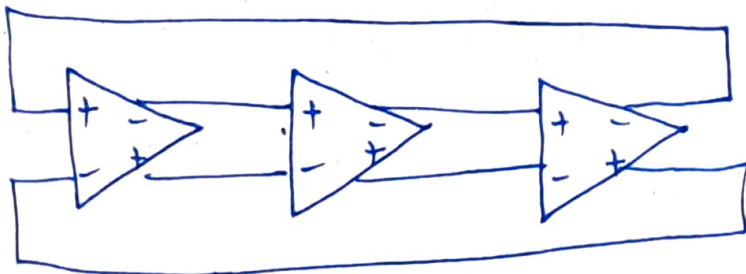
: at Common Source node means any current injected from one side to another will create significant voltage swing.

: if resistance is small.

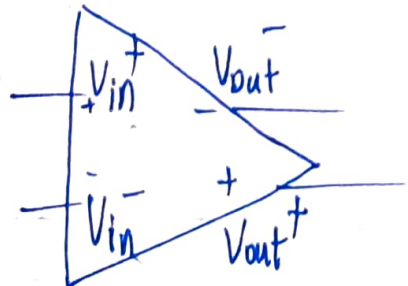
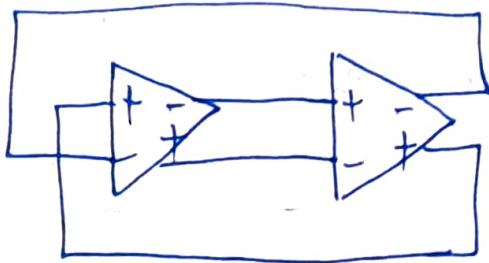
→ Injection is shunted to ground.

✓ Easy path into ground.

odd



even



Phase Consideration

4.2

Principal Drawback

:- Differential Rings have higher phase Noise.

①

Thermal phase Noise

$$S_{\phi_n} = \frac{8KT}{3I_{SS}} \left(\frac{\gamma}{V_{GS} - V_{th}} \right) + \frac{1}{I_{SS} R_D} \frac{f_0^2}{\Delta f^2}$$

inversely
 $\propto$
to tail current

Transistor
contribution

Load resistor
contrib.

Ex 4.3 * No. of Stages :

→ increasing N while power swing, ~~freq~~ constant

→ requires to reduce $(I_{SS}) \propto (W/L)$

→ S_{ϕ_n} rises as $1/I_{SS}$ rises. (Less N stages!!!)

②

Flicker noise Contributions

- More complex
- Arises when single-ended drain Voltage waveforms V_x, V_y have asymmetric rise & fall

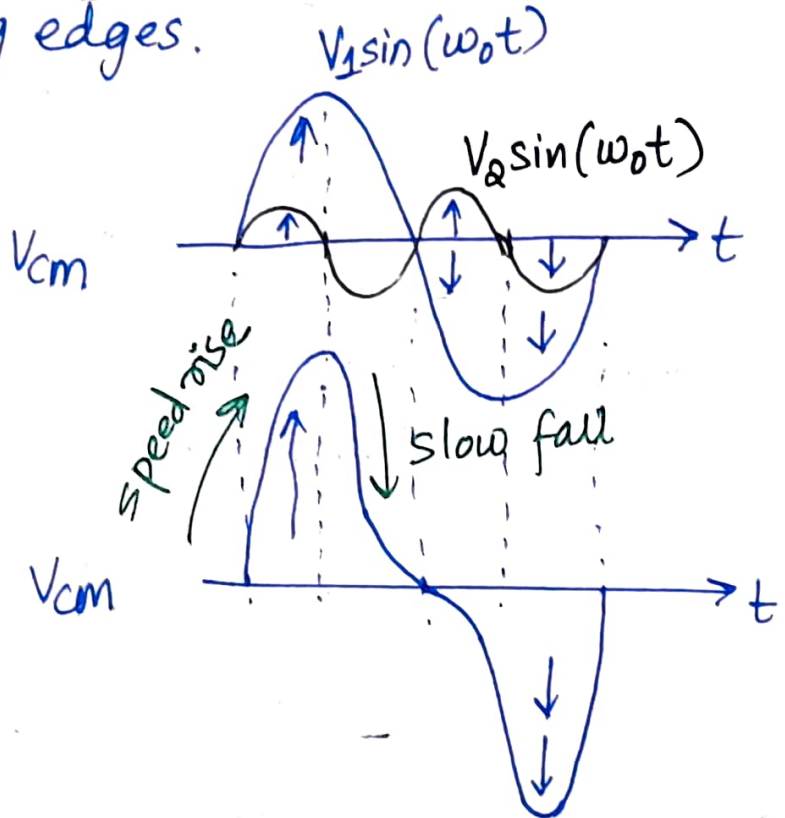
• Cause of Asymmetry? → Presence of 2nd harmonic in V_x & V_y .

2nd harmonics in V_x & V_y leads to unequal Slopes

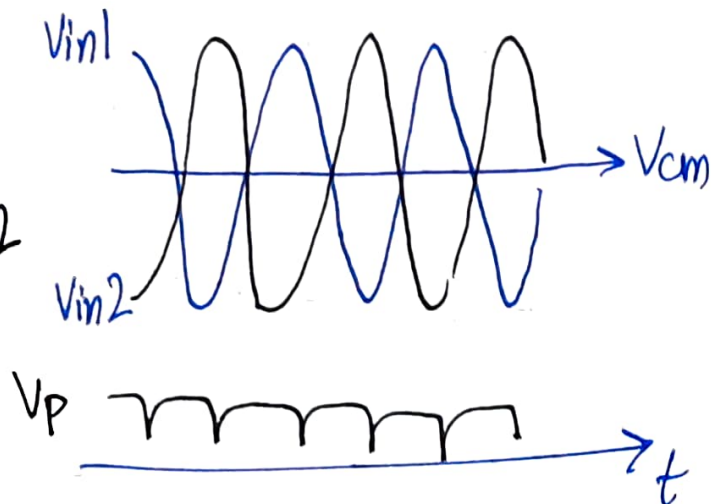
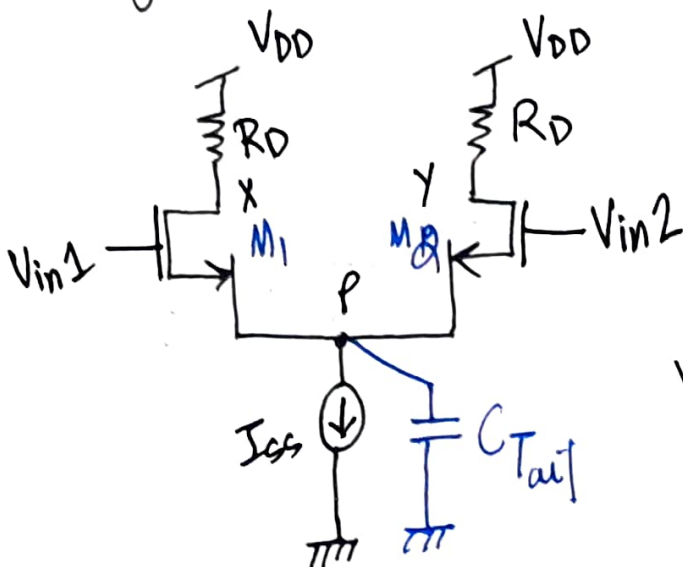
$$V_x(t) = V_1 \sin \omega_0 t + \underbrace{V_2 \sin(2\omega_0 t)}_{2^{\text{nd}} \text{ harmonic}} + V_{cm}$$

Example
4.4

can speed up rising edges & slow down falling edges.

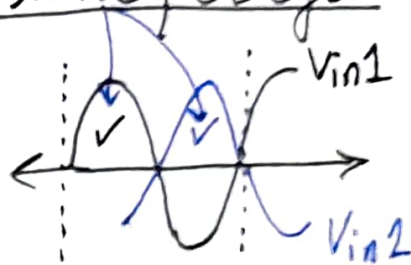


Origins of 2nd harmonics.



* V_p droops around crossing of V_{in1} & V_{in2}
 $\hookrightarrow$ Is due to switching of M_1/M_2 & C_T fluctuation

$\left(\begin{array}{l} \checkmark V_{in1} : \text{Max} \rightarrow M_2 \text{ off} \rightarrow M_1 \text{ Source follower} \\ \checkmark V_{in2} : \text{Max} \rightarrow M_1 \text{ off} \rightarrow M_2 \text{ Source follower} \end{array} \right) \rightarrow \text{lifting } V_p$
 V_p is lifted twice per cycle



* frequency of $V_p = \boxed{2\omega_0 = f_{VP}}$

* this $\boxed{2\omega_0}$ component in V_p generates current in C_T

increasing Transistor flows $\leftarrow$ through X & Y . $\leftarrow$ directly causing unequal slopes $\frac{C_T}{P_n V_x \& V_y}$
 widths

$\downarrow$
 increase 2nd harmonic $\leftarrow$ higher V_o ; larger C_T ; smaller $V_{gs} - V_{th}$.

* flicker Noise corner frequency (f_{1/f^3})

ratio of pos & neg slopes = α $\leftarrow$ asymmetry ratio
e.g., if $\alpha = \frac{v_1 + 2v_2}{v_1 - 2v_2}$

$$\rightarrow f_{1/f^3} = f_{1/f} \frac{3(1-\alpha)^3}{2N(1-\alpha+\alpha^2)}$$

Conclusion:

- * Larger asymmetry (α) increases corner frequency
- * flicker noise dominates over a width freq range.

Example 4.5 M_1 in prev diagram switches at the rate of f_0 .

If 2nd harmonic current drawn by C_T is mixed with f_0 ,

$$\begin{aligned} 2f_0 + f_0 &= 3f_0 \\ &\& 2f_0 - f_0 = f_0 \end{aligned}$$

Thus, tail-induced 2nd harm. does not ^{produced.} create asymmetry -

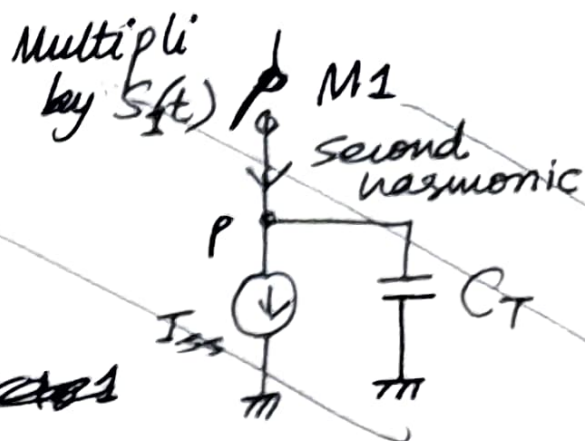
Wrong. Why?

Ans

M_1 acts as a Mixer

multiplies tail current by a square wave $S(t) \rightarrow$ ~~0 & 1~~

$\rightarrow$ toggling 0 & 1



* mix $2f_0$ with $f_0 \rightarrow f_0$ (diff.) $\Rightarrow$ student concluded $2f_0$ should not cause
 $\rightarrow 3f_0$ (add)

* ① M_1 is a mixer: V_{in} high $\rightarrow M_1$ on $\rightarrow$ asym. current flows
 $\rightarrow$ vice-versa.

② Switching $f^n S(t)$: $\rightarrow$ on/off action as square wave.
 $\rightarrow f_0$: oscillation of $S(t)$.

③ Fourier Sine of $S(t)$: $\rightarrow$ Broken down into series of sine wave.
 $\rightarrow$ DC component (a const. avg value)
 $\rightarrow$ fund. freq = f_0

④ Multiplication of $S(t)$: $\rightarrow$ odd harmonics ($3f_0, 5f_0, \dots$)

$I_{D1} \propto$ (Tail current, including $2f_0$ components) $\times S(t)$.

⑤ Critical point: DC Component Multiplies at $2f_0$.

$\rightarrow$ 0.5 from $S(t)$ multiplied by $2f_0 \rightarrow$ passes thro M_1 as I_{D1} .

$\therefore I_{D1}$ has $2f_0$ component $\Rightarrow V_x$ has it $\Rightarrow$ Asymmetry.

4.3 BASIC Differential Ring Design...

4.3.1 Initial design

- focus on performance & phase noise.

Requirements to ensure proper operation

① Start-up Condition: small-signal low-freq
Voltage gain of each stage atleast 2.

$$\boxed{g_m R_D \geq 2} \rightarrow \text{starts oscillation.} \leftarrow \begin{array}{l} \text{signal} \\ \text{can grow} \\ \text{\& start} \\ \text{osc.} \end{array}$$

$$\Rightarrow \frac{1}{V_{GS} - V_{TH}} \geq \frac{2}{R_D I_{SS}} \text{ is ensured.}$$

② Voltage Swings for Low Phase Noise :

- Bigger voltage swings help reduce phase noise (i.e., cleans signal)

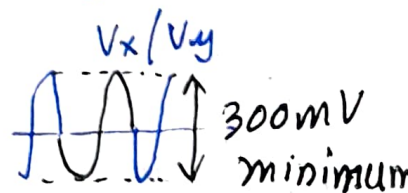
- A swing of $\sim 300\text{mV}$ is a good target.

- For full swing (max), current should switch fully b/w 2 MOSFETs

- then swing $V = I_{SS} \times R_D$

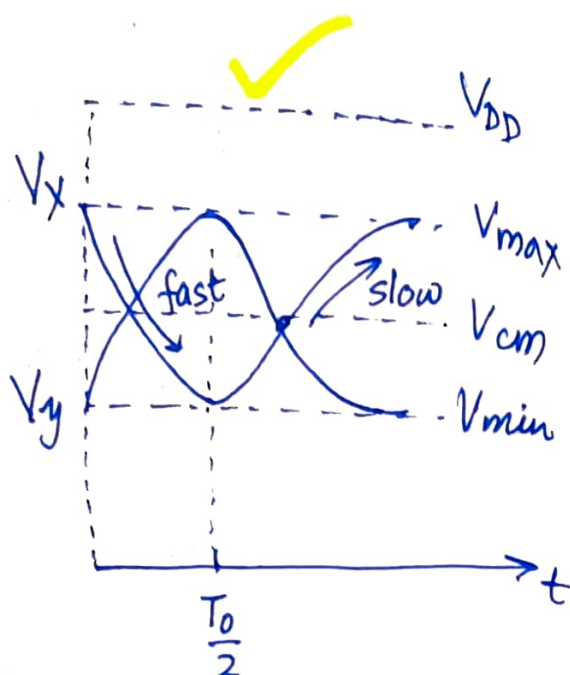
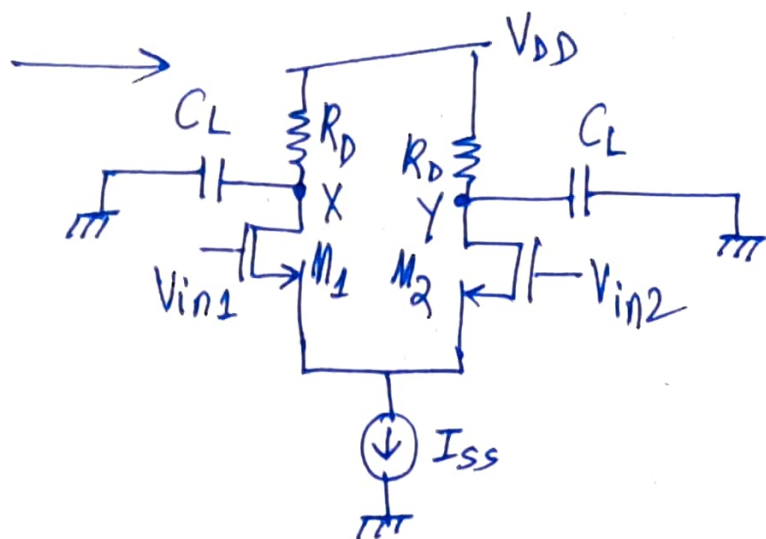
→ single-ended p-top swing of 300mV

(at one node X or Y)



Example 4.6

Prove that single-end voltage swings in a 3-stage ring cannot reach $I_{SS}R_D$ even with complete current steering.



osc period = T_0

for 3-stage ring $\Rightarrow T_0 = \frac{2\pi}{\sqrt{3}} R_D C_L = \frac{1}{f_0}$

Steps

M_1 : ON V_X is at Max; V_Y is zero

M_2 : is off C_L is getting charged by current

gradually turns on - this current is flowing from R_D

- V_X gradually reduces to zero

- V_Y increases

$T_0/2$ ← half oscillation

M_2 : ON V_Y is at Max; V_X is zero

M_1 : gradually off

V_y & V_x are complimentary

V_{min} & V_{max} $\rightarrow$ max & min values of o/p reaches.

V_{cm} $\rightarrow$ avg common-mode level of V_x & V_y swing

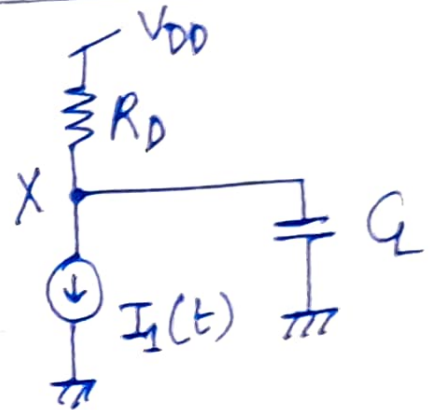
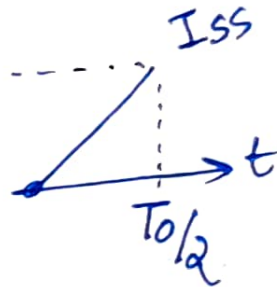
* Voltage swing is limited by how fast C charges & discharges through R_D .

* It never reaches V_{DD} ($V_{DD} = I_{SS} R_D$)

M_1 is ON: circuit becomes:

0 to I_{SS} in $T_0/2$ s

$$I_1(t) = I_{SS} \frac{t}{T_0/2} u(t)$$



$$V_x(t) = V_{max} - \frac{R_D I_{SS}}{T_0/2} \left\{ t + R_D C \exp\left\{ \frac{-t}{R_D C} \right\} - R_D C \right\}$$

$$V_x(0) = V_{max}$$

$$T_0 = 3.6 R_D C$$

$$\text{at } t = T_0/2 = \frac{\pi}{\sqrt{3}} R_D C = 1.8 R_D C$$

* current as ramp: $\rightarrow$ single-ended Voltage swing $\approx 0.54 I_{SS} R_D$

NOTE

① V_{cm} doesn't change

$$V_{cm} = V_{DD} - \frac{R_D I_{ss}}{2}$$

→ Balanced Resistors R_D on both sides → $I_1 = I_2 = \frac{I_{ss}}{2}$

$$→ V_{drop} = R_D \cdot \frac{I_{ss}}{2} \checkmark$$

$$V_y = V_x = V_{DD} - V_{drop} \quad \therefore V_{cm} = V_{DD} - \frac{R_D I_{ss}}{2}$$

↑
avg.

② Increase no. of inverters to reach $R_D I_{ss}$: full swing

→ more stages → less freq → more delay

better pull/push towards V_{DD} ← slow Transitions

③ Small-signal estimate of T_o still valid for large signal?

Yes! $\omega_o = \sqrt{\frac{3}{R_D C}} \rightarrow$ Based on 60° phase shift

large signals : higher I_{ss} , higher V_{max}/V_{min} $\checkmark 60^\circ$ is still req → Everything is SAME

Initial design Parameters & Simulation Results.

- $I_{SS} = 100\mu A$
 - $R_D = 6K\Omega$
- } single-ended swing $V_{out} \approx I_{SS} R_D$

$$V_{out} \approx 0.54 I_{SS} R_D \approx \boxed{320mV}$$

Target

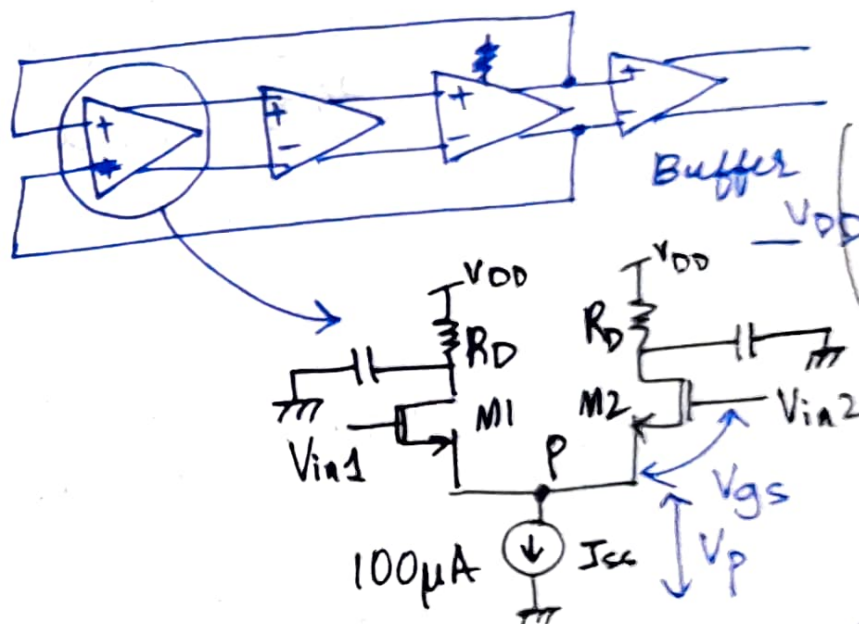
✓ Start-up condition

$$g_m R_D \geq 2$$

$$\rightarrow g_m \approx \frac{2 I_D}{V_{GS} - V_{TH}} = \frac{I_{SS}}{V_{GS} - V_{TH}}$$

$$a \quad \boxed{\frac{W}{L} = \frac{400nm}{40nm}}$$

✓ design



specs:

$$V_{DD} = 0.95V$$

$$R_D = 6K\Omega$$

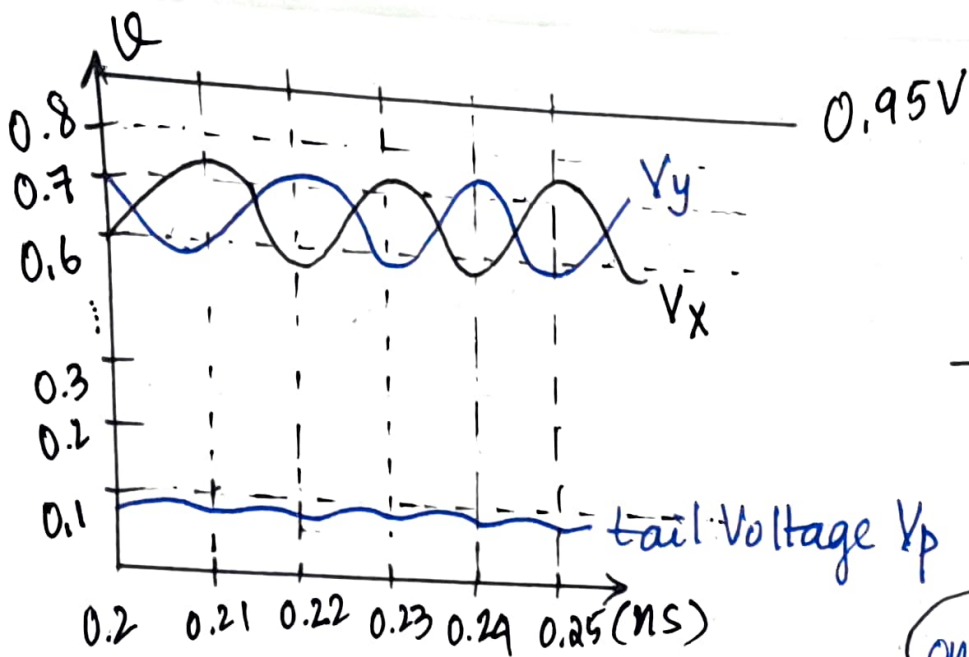
$$C_L = 0.6fF$$

$$\frac{W}{L} = \frac{400nm}{40nm}$$

$$I_{SS} = 100\mu A$$

$$T = 75^\circ$$

✓ SS corner



why?

→ due to short delay around loop

only about 150mV
($0.25 I_{SS} R_D$)

• Actual Single-Ended Swings

↓ due to incomplete steering of I_{SS}
calculated was 320mV; this is less
($\frac{I_{D1}}{I_{D2}}$ is b/w 30μA & 70μA instead of 0 → 100μA)
(By simulation)

• Oscillation freq: 42 GHz — twice as inverter rings. (it was 22 GHz)

• $V_p \approx 8mV$ tail Voltage: Low

→ due to large transistor overdrive

$$V_{GS} - V_{th} \approx 0.3V$$

→ leaving limited headroom for tail current source.

* What is headroom? what is trans overdrive?
why this phenomenon?

→ the tail current (typically NMOS)
needs minimum V_{DS} to be in SR

→ if V_p is very high, M_1 & M_2 may leave SR

→ headroom = $V_{DD} - V_p$

→ $V_p = 80\text{mV}$ & $V_{DD} = 0.95\text{V}$

∴ headroom = 0.87V

very low

harder to maintain stable I_{SS}

what is Overdrive?

→ $V_{OV} = V_{GS} - V_{th}$

→ how strongly transistor is turned on?

→ Larger overdrive = higher current drive.

→ But, it reduces V_{DS} headroom

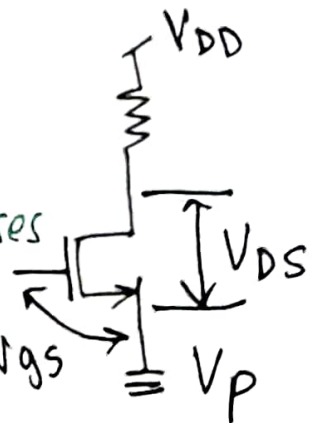
✓ to keep $V_{GS} = V_{OV} + V_{th}$: $V_{GS} = V_{DD} - V_p$: NMOS tail

So larger V_{OV} → smaller V_{DS} of tail

$V_p = 8\text{mV}$ ← less headroom
overdrive $V_{OV} = 0.3\text{V}$ ← larger

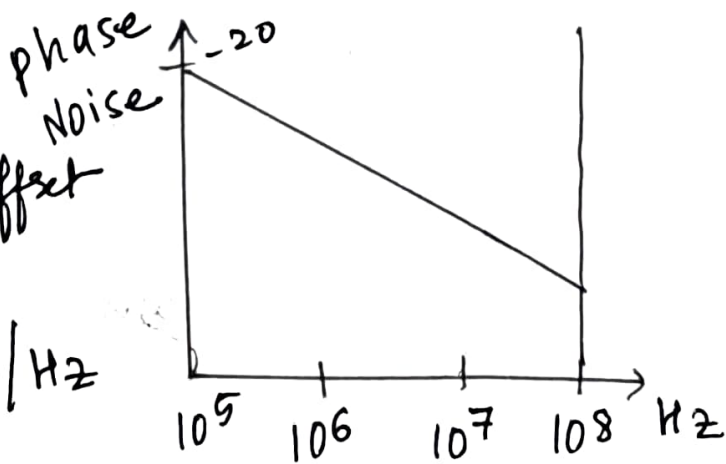
increases V_{OV} → V_p increases → V_{DS} decreases

$V_p = V_s = V_{GS} - V_{GS}$ (in small-sig $V_g \rightarrow \text{fixed}$)
 $V_p \propto V_{GS}$



Phase Noise Analysis (4.11)

- flicker Noise dominates below $\sim 1\text{MHz}$ offset



- At 100MHz offset
 $S_{\phi_n} = -90\text{dBc/Hz}$
thermal noise

Prediction

- $T = 75^\circ\text{C}$, $I_{ss} = 100\mu\text{A}$; $V_{gs} - V_{th} = 300\text{mV}$
 $\rightarrow S_{\phi_n} \sim -99.5\text{dBc/Hz}$

[using S_{ϕ_n} eqⁿ discussed before]

* Simulated is higher than Calculated

- $\rightarrow$ Only $0.25 I_{ss} R_D$ swing observed
- $\rightarrow$ NO complete current switching.

- upconversion: Low freq noise ($1/f$) is converted to phase noise at higher frequency near osc o/p frequencies

* this design has ^{less} $(1/f)$ noise upconversion than single-ended rings.

- $\rightarrow$ Small swing
- $\rightarrow$ large $V_{gs} - V_{th}$.

Performance Comparison

diff

- $f = 42 \text{ GHz}$
- S_{ϕ_n} at $100 \text{ MHz} = -90 \text{ dB/Hz}$
- $P = 3 \times 100 \mu\text{A} \times 0.95 \text{ V}$
 $= 285 \mu\text{W}$
- FOM $\rightarrow$ less

Single-ended

- $f = 22.6 \text{ GHz}$
- S_{ϕ_n} at $100 \text{ MHz} = -104 \text{ dB/Hz}$
- $P = 3 \times 60 \mu\text{A} \times 0.95 \text{ V}$
 $= 57 \mu\text{W}$
- FOM: 15.6 dB better

4.3.2

Design Improvements

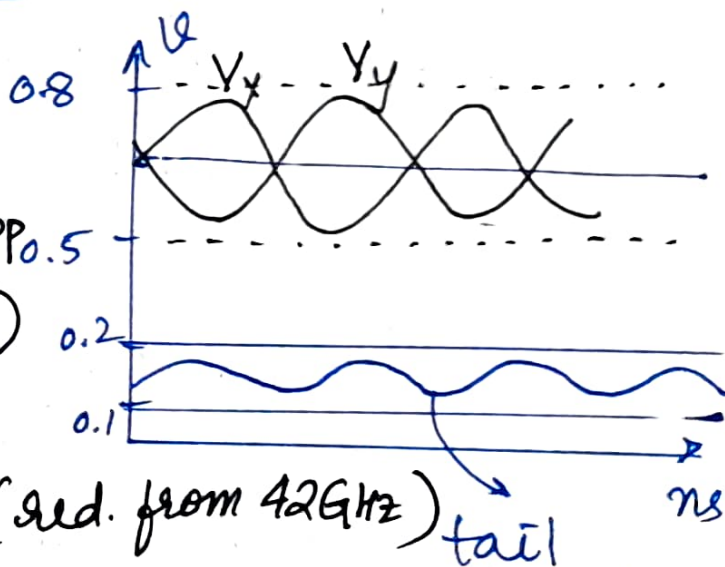
- ✓ increase W
- ✓ performance.

Improvement 1: increase the width. (double them)

$$W/L = \frac{800 \text{ nm}}{40 \text{ nm}} \leftarrow \text{to } 400 \text{ nm}$$

- V_x, V_y increase to 250 mV_{PP}

drain current
swing by (from 150 mV_{PP})
 $17.5 \mu\text{A}$ to $87 \mu\text{A}$



- OSC freq = 31.3 GHz (red. from 42 GHz)
- V_p swing from to $\sim 35 \text{ mV}_{PP}$
 $\Rightarrow$ leading to higher $2f_0$ current due to tail capacitance.

- * higher displacement current at 2nd harmonic in the tail capacitance. [due to higher tail ~~current~~ volt. swing]
 - * Second harmonic in drain voltages becomes -23dB (from -37dB)
- increased Asymmetry.

what????

- V_p increases (swing $\uparrow$)
- this swing at p interacts with tail capacitances.
- generates current at $2f_o$ [2nd harmonic of osc freq]
- this current flows thro diff pairs distorting drain voltages
- ⇒ becomes dominant [-23dB $\leftarrow$ -37dB]

Phase Noise

Thermal : 100MHz offset

✓ drop ~ 8 dB

✓ Expected ~ 7 dB
(2.6dB from freq
4.4dB from swing)

S_{pn} predicts -101dBc/Hz

$V_{gs} - V_{th} = 220$ mV

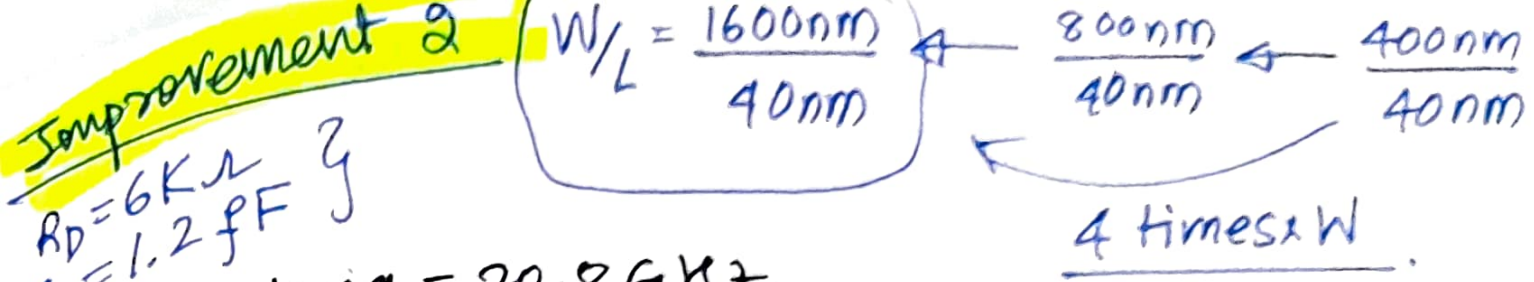
4dB lower than simulated

less symmetry & less linear

Flicker : Expected 7dB drop (thermal) + 3dB drop (doubling area)

✓ observed = 7dB only. $\leftarrow$ due to higher 2nd harmonic & inc. asymmetry

(Worsen Flicker noise upconversion)



$R_D = 6\text{K}\Omega$
 $C_L = 1.2\text{fF}$

osc freq = 20.8 GHz

V_X, V_Y slightly increased to 270mVpp.
 (closer to $0.54 I_{SS} R_D$ value)

- 2nd harmonic in drain voltage } more increase by -19 dB } asymmetry
- Rise & fall times remain asymmetric
 21ps 28ps

Phase Noise : $\nearrow$ Thermal

- Expected : $\sim 4.25\text{dB}$ $\nearrow$ 3.55 from freq
- Actual : 4.5dB $\nearrow$ 0.7 from swing

S_{ϕ_n} predicts -103.7 dBc/Hz
 1.2dB lower than simulated
 $\rightarrow$ closer to ideal

flicker

- Expected : 7.25dB (thermal + area)
- Actual : 6dB reduction

} due to higher 2nd harmonic & increased asy.

- Conclusion :
- ① Increase W to increase Voltage swings and reduce thermal phase noise
 - ② But it lowers Osc freq; exacerbates 2nd harmonics $\rightarrow$ worsen flicker noise.
 - ③ these consume more power, worse thermal ph noise upconversion

4.4 Obtaining the desired frequency

aim $\Rightarrow f_0 \approx 2 \text{ GHz}$

① Add cap

② increase length & width

③ increase N

{ dividing frequency doesn't improve FOM }

4.4.1 Adding capacitance

$$R_D = 6 \text{ k}\Omega$$

$$f_0 = 20.8 \text{ GHz}$$

$$W/L = \frac{1600 \text{ nm}}{40 \text{ nm}}$$

$$C_L = 2 \text{ fF}$$

$$20.8 \text{ GHz} \xrightarrow{\text{to}} 2 \text{ GHz}$$

approach: inc C_L to lower f_{osc}

$$f_0 = \frac{\sqrt{3}}{2\pi R_D C_L}$$

$$T_0 = \frac{2\pi}{\sqrt{3}} R_D C_L$$

remember?

osc formula for 3-stg diff ring $\rightarrow$ estimated $C_L = 2.21 \text{ fF}$

- $20.8 \text{ GHz} \rightarrow 2 \text{ GHz}$: drop by factor of ~ 10
- increase C_L by factor of 10

$$\therefore C_L = 25 \text{ fF} \longrightarrow \text{yields } f_0 = 2.14 \text{ GHz}$$

• Phase Noise

→ Thermal Noise

- theoretical

$$10 \log \left(\frac{20.8}{2.14} \right)^2 = 19.8 \text{ dB}$$

- Actual $\Rightarrow \sim 18 \text{ dB drop}$

✓ lower phase noise as osc freq $\downarrow \downarrow$

→ Flicker Noise : actual is 18 dB drop,
✓ lower flicker noise as $C_L \uparrow$ & osc $\downarrow$ freq

-
- K_{VDD} : ✓ 74 MHz/V Σ it was 1.82 GHz/V for 20.8 GHz circuit
 - ✓ significant benefit

(why?) $\rightarrow$ 25 fF Capacitance is larger than transistor's ^{extrinsic} intrinsic cap C_{DB}

✓ total node cap is dominated by C_L and less dependent on $C_{DB} \rightarrow$ less supply V_{DD} sensitivity

-
- Normalised K_{VDD} : 2-5 times better than before.

4.4.2 Method 2 : Larger Transistors.

$$\frac{W}{L} = \frac{1600\text{nm}}{40\text{nm}} \longrightarrow \frac{8\mu\text{m}}{240\text{nm}}$$

$$f_0 = 2.27 \text{ GHz}$$

- width is increased as well to get enough headroom for V_p

$$\begin{aligned} V_{gs} - V_{th} \\ \text{headroom} = V_{DD} - V_p \\ V_p \propto V_{gs} \end{aligned}$$

- interconnect Cap $\longrightarrow$ 2-4 fF from 1.2 fF for larger cell size & longer wiring.

- osc freq = 2.27 GHz

- $R_D = 6 \text{ k}\Omega$

$$K_{VDD} = 228 \text{ MHz/V}$$

① increased Voltage Swings.

- V_x & V_y increased ~~to~~: 270mV to 360mV
- very close to $0.54 I_{SS} R_D$ swing
- ✓ complete switching

② Phase Noise

* Thermal phase noise $\longrightarrow$ drop by 21.5dB

- observed = 21.5dB

- calculated = 22dB

$$20 \log \left(\frac{20.8}{2.27} \right)$$

$$= 19.5$$

$$20 \log \left(\frac{360}{270} \right)$$

$$= 2.5$$

* flicker noise: by 26dB drop

Expected due to scaling = -19.2dB

↳ addⁿ due to inc area = 15dB

Actual: $\sqrt{26} - 19.2 = 6.8$ → very less than expected 15 from area.

* flicker noise: drop by 26dB.

• Expected: → due to scaling = -19.2dB

→ due to inc area = 15dB

($10 \log(\text{Area Ratio})$)

$$\rightarrow \frac{8000\text{nm} \times 236\text{nm}}{1600\text{nm} \times 36\text{nm}}$$

• Actual: $26\text{dB} - 19.2\text{dB} = \boxed{6.8\text{dB}}$ is less than 15dB expected.

Why?

large Transistors }
Incr. swings

lead to inc cap value
(tail cap)

✓ rise in 2nd harmonic in V_x & V_y to -12dB.

→ degrades flicker noise. ← asymmetry introduced.

Larger transistor is superior in phase Noise

K_{vpd} little worse than ①

4.4.3 Greater No. of Stages

increase $N \rightarrow$ lowers fosc.

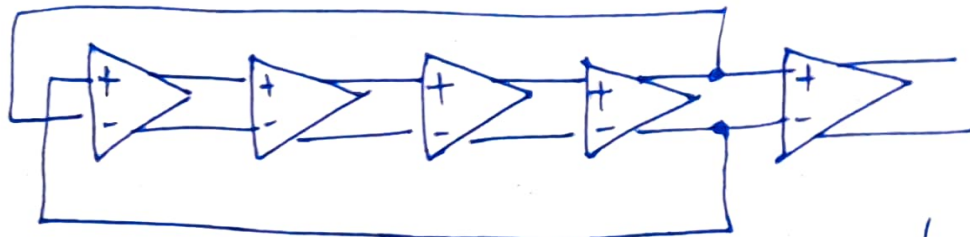
two opposing trends:

① higher $N \rightarrow$ higher phase noise
(for const. power)

② higher $N \rightarrow$ Larger swings $\rightarrow$ Lower Normalised
Phase Noise
(3 to 4 to 5)

- voltage swings increase
- lower phase noise

✓ from 3 to 4 ✓ moderate



$$\frac{W}{L} = \frac{4\mu\text{m}}{240\text{nm}}$$

$$C_L = 1.2\text{fF}$$

$$R_D = 6\text{k}\Omega$$

$$N = 4$$

• I_{SS} (tail current) = $75\mu\text{A}$
(red. from $100\mu\text{A}$)

• Requires a cross over in feedback lines for neg feedback.

* Voltage Swings : reaches 420mV_{pp}

* OSC freq : 2.16GHz

* KVDD : $194\text{MHz}/V$

* Phase Noise $\rightarrow$ 1dB higher than "Larger Tran" expt.

$\rightarrow$ two opp. trends cancel each other out at 3 to 4 stages: N.

Overall

• Best FOM : Larger Trans.

• Four-stage Advantage :

✓ Larger Volt. Swings

✓ More o/p's Phases

4.5 Two Stage Ring Osc

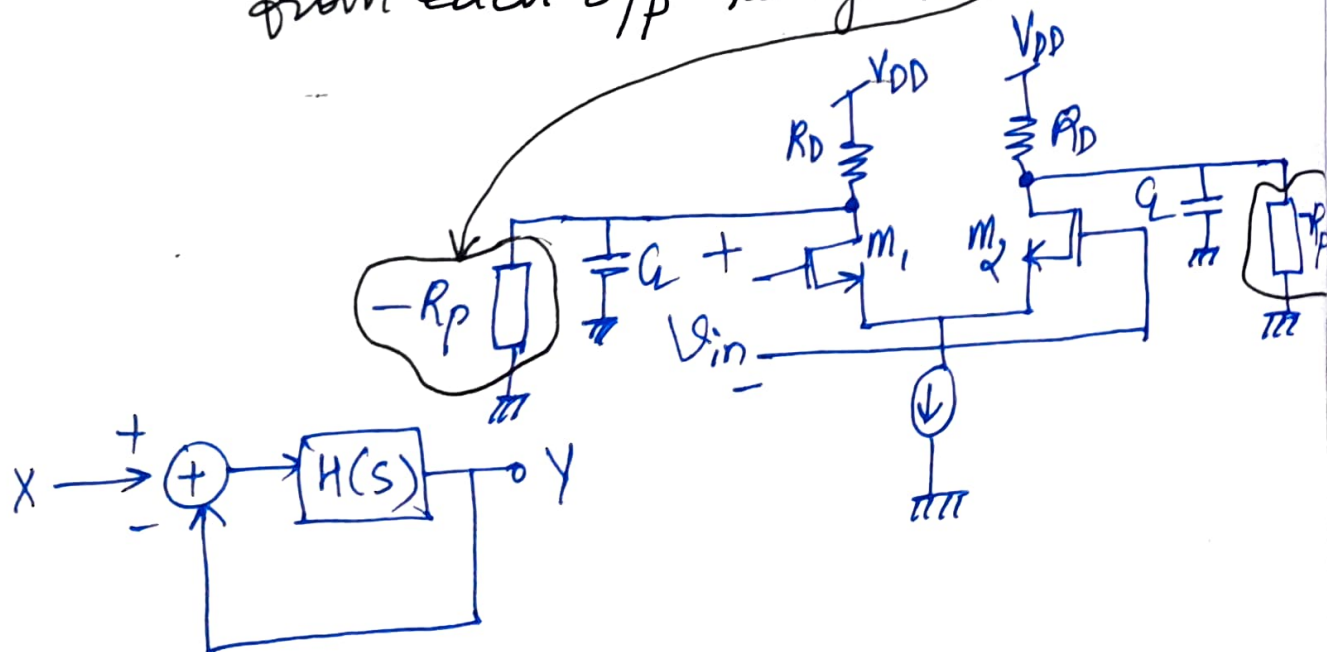
4.5.1 Basic Idea

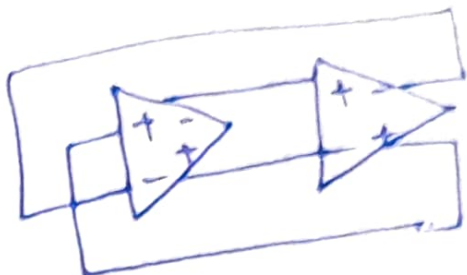
* Initial Challenge : two simple diff pairs cannot oscillate on its own.

- forms a loop with only 2 poles
- doesn't provide 180° phase shift at any finite frequency

* Proposed Solution Negative Resistance Concept

- To enable Oscillation in a 2-stage ring
- each stage is conceived as a diff pair with a neg resistance $-R_p$ connected from each o/p to gnd.





Transfer function

- TF of a single stage shows that if $R_p < R_D$

$$\frac{V_{out}}{V_{in}} = -g_{m1,2} \left[R_D \parallel \frac{1}{C_L s} \parallel (-R_p) \right]$$

the parallel comb of R_D & $-R_p$ becomes negative

$$\frac{V_{out}}{V_{in}} = -g_{m1,2} \frac{-R_D R_p}{(R_D - R_p)} \frac{1}{\frac{-R_D R_p}{R_D - R_p} C_L s + 1}$$

of 1st stage.

- R_D & $-R_p$ parallel combo is effectively negative

$$R_L = \frac{R_D R_p}{R_D - R_p} \quad \left\{ \begin{array}{l} \text{where} \\ R_L > 0 \end{array} \right.$$

* Loop Transmission Analysis

$H(s)$ is loop transmission, is a square of the single-stage T-F:

$$H(s) = \left(\frac{V_{out}}{V_{in}} \right)^2$$

- To find osc condition $\left\{ \begin{array}{l} \text{closed loop TF} \rightarrow \infty \text{ when} \\ H(s) = -1 \end{array} \right.$

solve for 's' = $\frac{1}{R_L C_L} \pm j \frac{g_{m1,2}}{C_L}$

2 complex poles in the right half-plane.

Soln to this is $\exp(\sigma_1 \pm j\omega_1)$ sinusoid at ω_1 but grows exponentially.

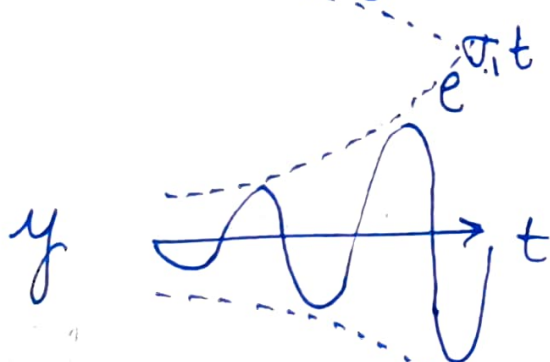
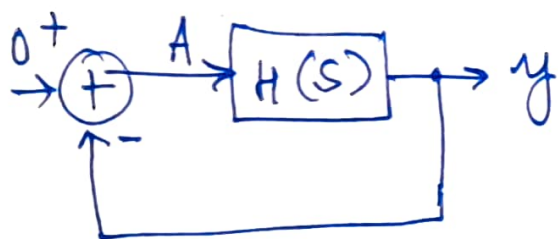
Implication of RHP poles

- ? circuit is unstable
- ? generates a sinusoidal wave that grows exponentially.
- ? grows till circuit non-linearities stop the amplitude

$$s = \sigma_1 \pm j\omega_1$$

(or saturation of neg resistance or differential pairs)

Example 4.7



Example 4.7

- If $R_p = \infty$ (no neg resistance) $\rightarrow s = \frac{1}{R_D C_L} \pm j \frac{g_{m1,2}}{C_L}$
 - $\rightarrow R_L = -R_D \rightarrow$ poles on left side
 - $\rightarrow$ no oscillations
 - $\rightarrow$ confirms the 1st challenge.

• ~~$R_p = R_D$~~

also true if $R_p > R_D$

If $R_p = R_D \rightarrow R_L = \infty \angle \phi = \pm j \frac{g_{m1,2}}{C_L}$

• $R_p = R_D$: load resistance for NMOS & PMOS is same

$R_L \rightarrow \infty$: tail current source is ideal

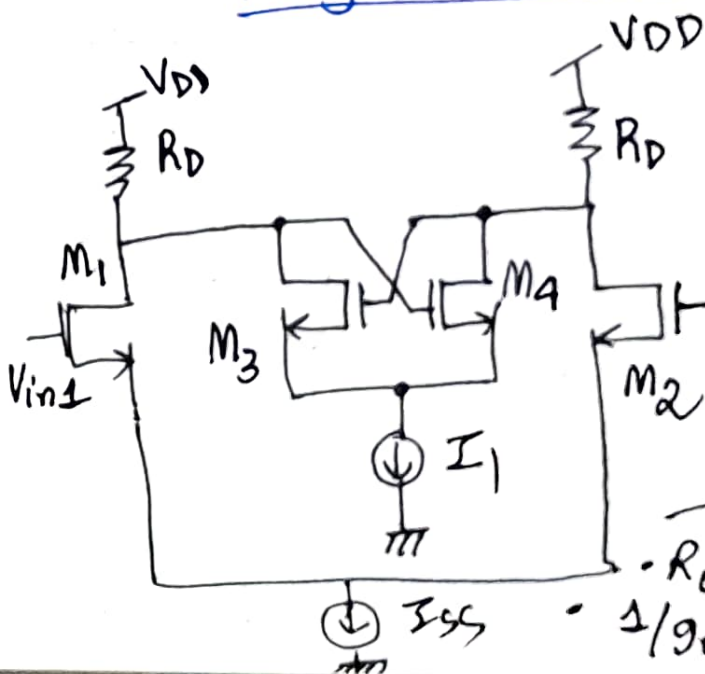
Ring reduces to two ideal integrators
at $\omega = g_{m1,2}/C_L$

• Integrators: takes an p/p voltage & slowly builds up (or reduces) the output voltage over time.

Rate of change of o/p $\propto$ rate change in input.

✓ perfect osci
✓ no delay/decay.

Practical Realisation of Neg Resistance



cross-coupled pair

M_3 & M_4 introduce a resistance equal to $-2/g_{m3,4}$

b/w X & Y
 $\Rightarrow R_p = 1/g_{m3,4}$

Condⁿ for oscillation.

R_L overall effective res to be positive

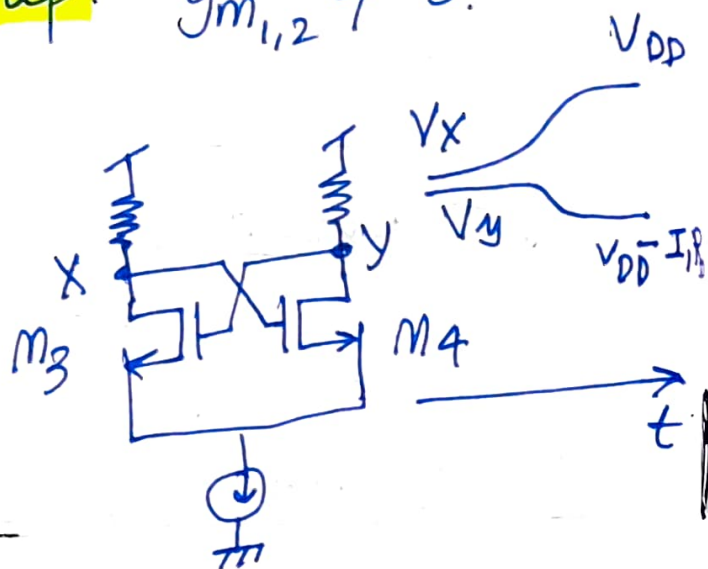
$1/g_{m3,4} < R_D ; g_{m3,4} R_D > 1$

Example 4.8 Can $g_{m1,2}$ be chosen arbitrarily

* If $g_{m1,2} = 0$ → $M1/M2$ diff pair reduces to 2-uncoupled cross-coupled pairs
 [$M3/M4$ → act as neg ses.]

→ these alone can't oscillate.

→ Latch up. $g_{m1,2} \neq 0$.



* Latch-up - problem

→ $g_{m3,4} R_D$ must be > 1 for poles to be created on right-hand side.

→ positive feedback in cross-coupled circuit around the loop $V_X \& V_Y$ pushes $V_X \& V_Y$ in opp directions

→ loop gain, $(g_{m3,4} R_D)^2 > 1$

→ forcing $V_X \& V_Y$ to diverge completely until one MOSFET switches off and the other carries all the current

* M3 & M4 in cross-coupled.

- ↳ these "boost" any imbalance.
- ↳ if say M3 gets more current,
 - it pulls its node down,
 - it raises gate voltage of M4
 - one MOSFET is off now,
 - another one conducts all current

∴ $V_{DD} = \text{one mode \{say } V_x\} \text{ output}$] **STUCK.**
 $V_{DD} - I_{SS} R_D \text{ \{other, } V_y\}$] **No Osc!**

- **no negative resistance to help oscillate.**

Solution

- Main pair M1 & M2

So to have negative resistance & oscillate!

→ to inject enough differential current into the circuit

→ periodically kick the latched pair (M3 & M4) back to active mode (SR)

→ $g_{m1,2}$ large ; I_{SS} large

more $g_{m1,2}$ → stronger control over o/p
→ input changes create current changes.
strong

more I_{SS} → total current for M_1 & M_2
→ more 'kick' / 'push' to force
MS & M4 out of latch-up.
create changes to their "fixed voltages"
to → active region

Advantages for 2-stage

- ① higher Osc freq.
- ② Quadrature O/p's: 4 o/p phases; 90° apart

Disadvantage

- ① Smaller Swings
→ due to short delay

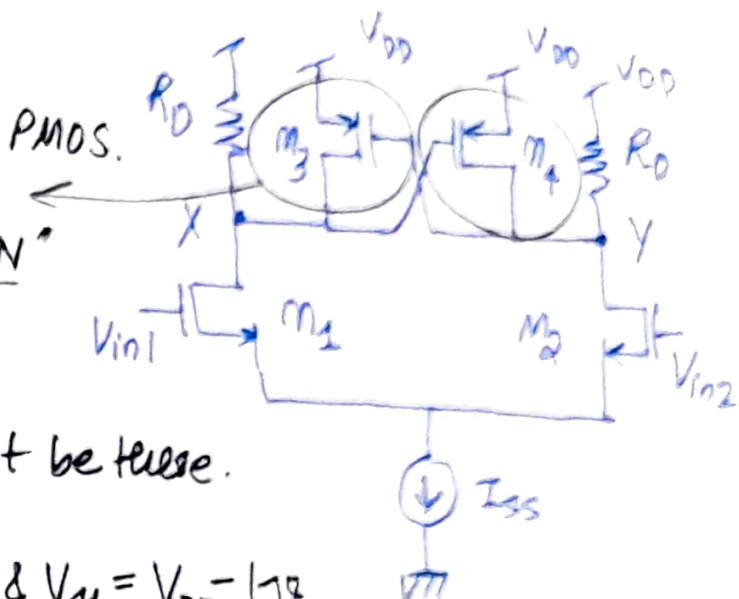
Example 4.9

PMOS cross coupled pair

- PMOS M_3, M_4 draw current from V_{DD} through R_D

- for M_3 & M_4 to be "ON" providing & neg res

$|V_{GS3,4}|$ must be there.



- Common mode : V_x & $V_y = V_{DD} - |V_{GS3,4}|$
- V_{DD} need to accommodate → Voltage drop across PMOS $V_{GS3,4}$



$$V_{DD}(\text{PMOS Case}) \geq |V_{GS3,4}| + |V_{DS1,2}| + V_{ISS}$$

Comparison to NMOS

- I_1 flows out of common ^{source} node of M_1 & M_2
- $I_1 + I_{SS}$ flows out of resistor.
- Voltage across R_D is due to $(I_1 + I_{SS})$ or ^{fraction} of it.
- Headroom Constr on V_{DD} involves R_D drops plus $V_{DS1,2}$ & V_{ISS} .

$$V_{DD}(\text{NMOS}) \geq \frac{(I_{SS} + I_1) \cdot R_D}{2} + V_{DS1,2} + V_{ISS}$$

✓ assumes an average current across R_D .

Why is PMOS Worse for headroom?

→ PMOS adds $|V_{gs3,4}|$ to V_{DD} drops

→ this is a fixed threshold-like voltage which has to be always present.

→ in NMOS: I_1 flows thro R_D doesn't impose a direct V_{gs} voltage drop in series with the main diff pair & tail curr source to det. V_{DD}

* headroom is more about currents which cause drop across R_D .

NOTE

- PMOS : it forces V_{DD} to be higher
- consumes more headroom

Example 4.10 A simple 2-stage diff ring will latch-up due to the feedback and why it is wrong?

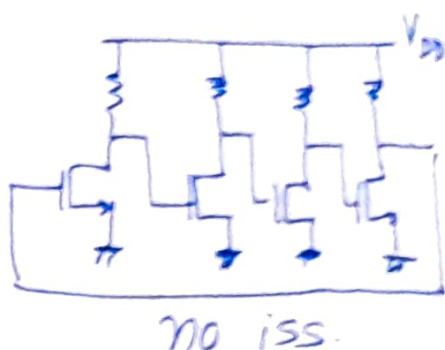
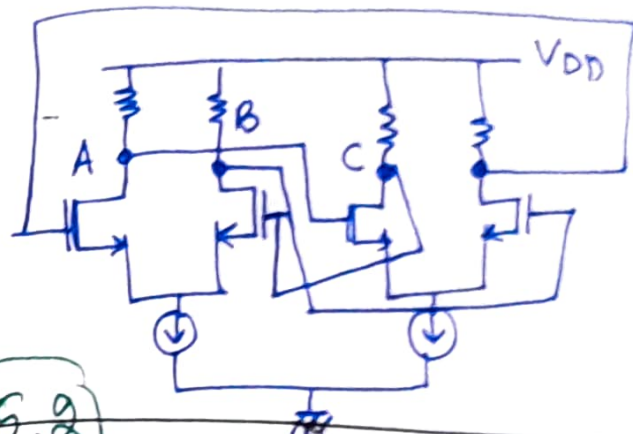
→ why it doesn't latch up?

- ① Anti-phase inputs due to diff operation.
- ② Impossible Symmetric Latch-up: student proposed $A=B=\text{high}$; $C=D=\text{low}$ is impossible

for differential pair. It implies M_1, M_2 off & M_3, M_4 on.

③ Tail Current Source's Role : I_{SS} steer current
- force them to be in active region.

④ Say no I_{SS} : it would latch up.



4.5.2

Design Example

① Initial Design

$$R_D = 4K\Omega$$

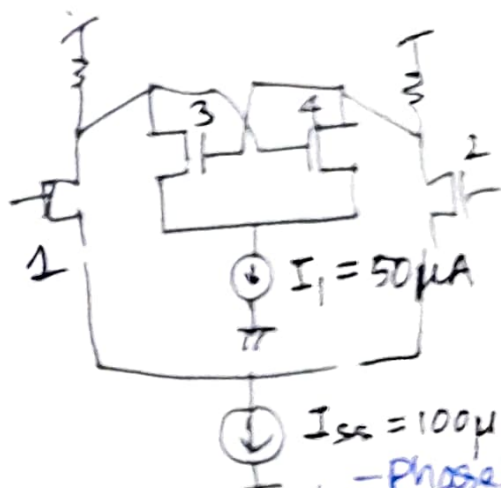
$$C_L = 1.2fF$$

$$(W/L)_{1,2} = \frac{8\mu m}{240nm}$$

$$(W/L)_{3,4} = \frac{4\mu m}{120nm}$$

Goal: to get 2GHz & $P = 300\mu A \times 0.95$

$$I_{total} = 100\mu A + 50\mu A$$



- Phase Noise
• 3 dB to 4 dB
higher than
3-stage.

Reason —
- small
swings.

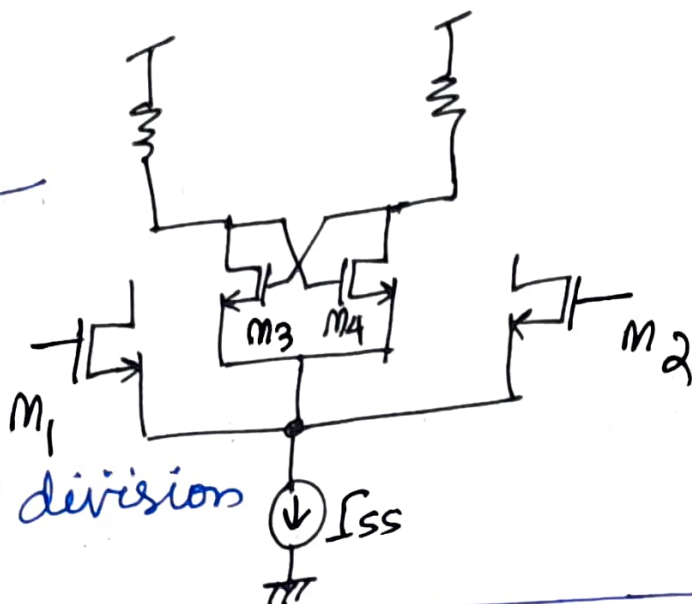
Simulation Results: $f_0 = 2.14GHz$; Swings: Less than 250mVpp
(3-stage has 300mVpp)

II Reconfig design

Shared tails

Current splitting.

- all transistor lengths must be equal for proper I division



- Diff pair :
$$\frac{I_{SS} \cdot W_{1,2}}{(W_{1,2} + W_{3,4})}$$

- Cross-coupled :
$$\frac{I_{SS} W_{3,4}}{(W_{1,2} + W_{3,4})}$$

Performance :

- Voltage Swings : Large swings observed.
- Flicker Noise : 10-15 dB higher fl. noise upconversion

Example 4.11 tail node waveforms of diff & cross-coupled pairs.

of one stage : tail nodes are off by 90° input/output

✓ seperate tails

- P generates ω_0 voltage wavef
- Q node, receives 90° shifted o/p
- V_P & V_Q are 180° out of phase
- also generates ω_0 V wavef.

✓ shared tails

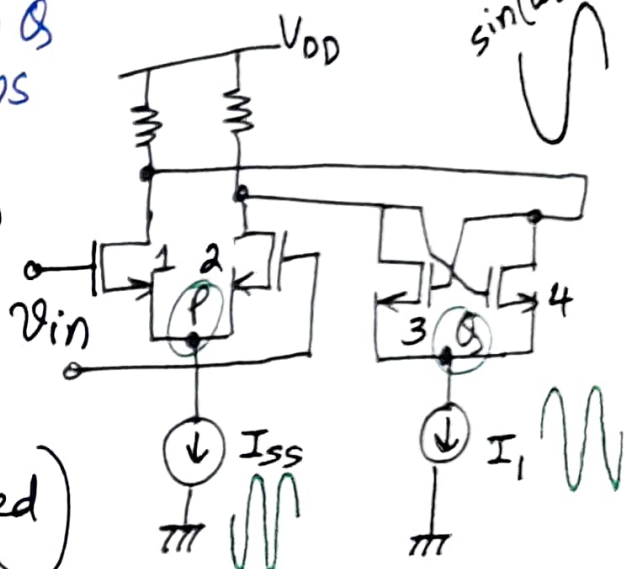
- P & Q are shorted. 180° out of phase ω_0 waveforms partially cancel each other.
- If they cancel Exactly.

the shared P & Q node voltage remains **SAME**.

- this looks as an **AC ground**.

(voltage is constant but it behaves grounded for high frequencies)

$\sin(\omega_0 t)$



- Increased 2nd order non-linearity & flicker noise.

→ due to absence of comm. source node voltage

→ since is AC grounded. (V_P)

- asymmetry & higher f-noise upconversion.

NOTE: this occurs even without tail capacitance

4.6 Linear Tuning

* Phase Noise of any oscillator can be reduced by exploiting its fundamental trade-off with power dissipation.

4.7 Tuning Techniques

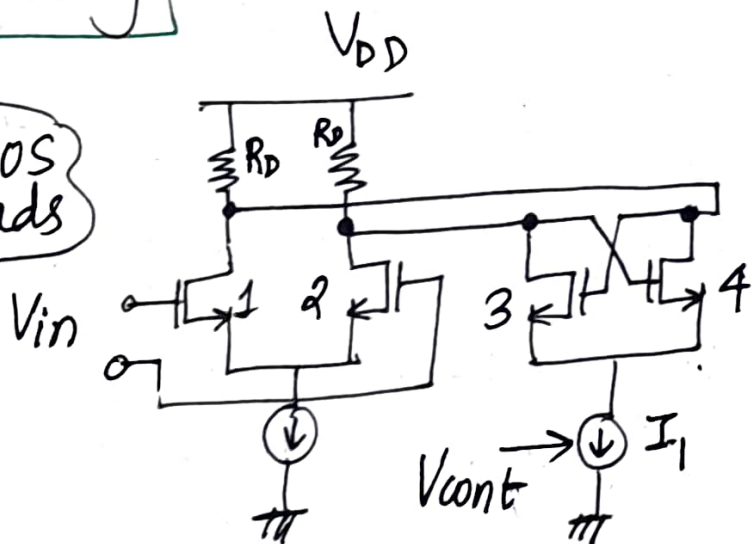
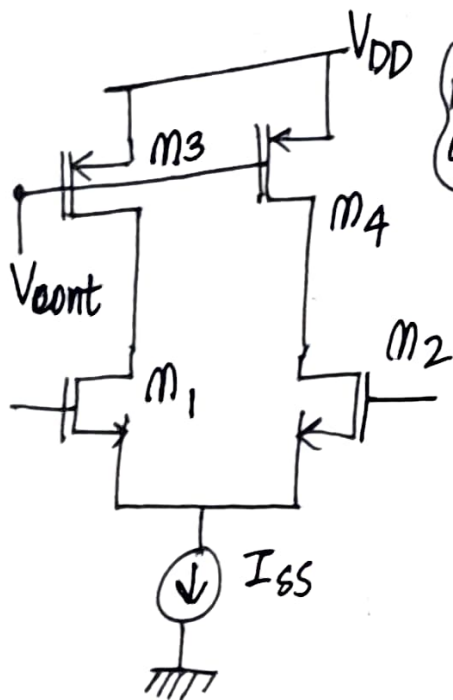
- Take larger transistor Example

$$\omega_0 = \frac{1}{R_D C_L} \tan\left(\frac{180^\circ}{N}\right)$$

Resistor
Cap
N

changes can be done.

4.7.1 Resistive Tuning



① PMOS loads

- Replace R_D with PMOS op in deep-triode region.
- these act as variable resistors cont. by V_{cont} .

✓ Problems

① Voltage-Swings change: As PMOS resistance varies, voltage swings at the o/p also change.
→ phase noise is affected.

② Limited tuning & Non-linearity:
Range

V_{cont} has a limited range to tune.
caz PMOS needs to be in triode.

✓ high non-linear freq tuning

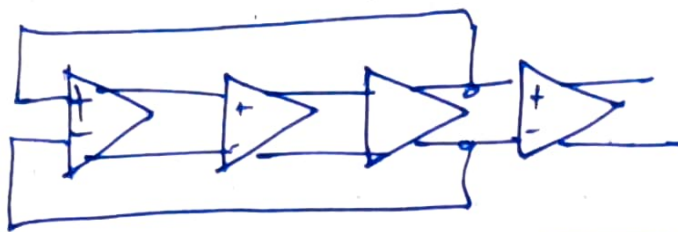
Resistance vs V_{cont} is Non-linear
so freq vs V_{cont} becomes

② Neg vari Resistor

- variable neg res created by M3 & M4 in cross-coupled in parallel with R_D in diff circuit
- I_1 current sets the strength of neg res

Advantage: can pot. minimise phase noise
 → if tuning range is small ($10-20\%$ f_0)

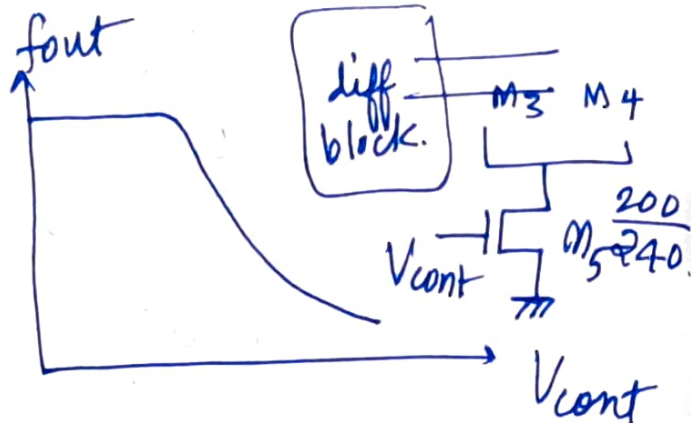
Example:



$$R_D = 6\text{K}\Omega$$

$$(W/L)_{1,2} = \frac{8\mu\text{m}}{240\text{nm}}$$

$$(W/L)_{3,4} = \frac{400\text{nm}}{240\text{nm}}$$



✓ Voltage swings increase

$$350\text{mV} \longrightarrow 480\text{mV}$$

$$✓ f_0 = 2.2\text{GHz} \longrightarrow 1.83\text{GHz}$$

$$✓ V_{\text{cont}}: 250\text{mV} \longrightarrow 900\text{mV}$$

$$✓ K_{\text{VCO}} = 1.6\text{GHz/V} : \text{Maximum}$$

M_5 in SR

Example 4.12 if M_5 enters triode region

- bias current $I_{D3,4}$ becomes dep. on V_{DD}
- M_5 acts like a resistor than C.S
- V_{DD} fluctuations directly modulate $I_{D3,4}$.
- $I_{D3,4} \longrightarrow$ modulate $g_{m3,4}$.
- ✓ changes effective $(-ve)$ resistance

thereby modulating freq.

→ Supply rejection gets worse.

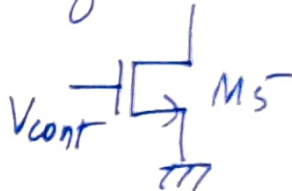
Phase Noise with tuning

- Thermal: f_0 : 2.2GHz to 1.83GHz
phase noise expected to drop.
1.6 dB ← freq scaling
Some more ← swings.
→ Actual 3dB drop

- Flicker: unchanged.

→ M_5 becomes contributors for phase noise
→ flicker noise directly modulates V_{cont} .

→ f_0 is changed



"flat zone": NMOS-only tuning

- NMOS+PMOS for a wider linear tuning
is Hard.

→ due to limited headroom for PMOS & low V_{DD} .

Overall

① Lower PVT variations ~~with no method~~ compared to invt rings.

② Why????

→ f depends on R_D & C_L → More stable than transistors

→ $\pm 15\%$ for R_D
 $\pm 5\%$ for C_L } for $f \pm 20\%$. ✓

4.7.2 Varactor Tuning

• MOS varactors

to vary $C \rightarrow$
tune freq

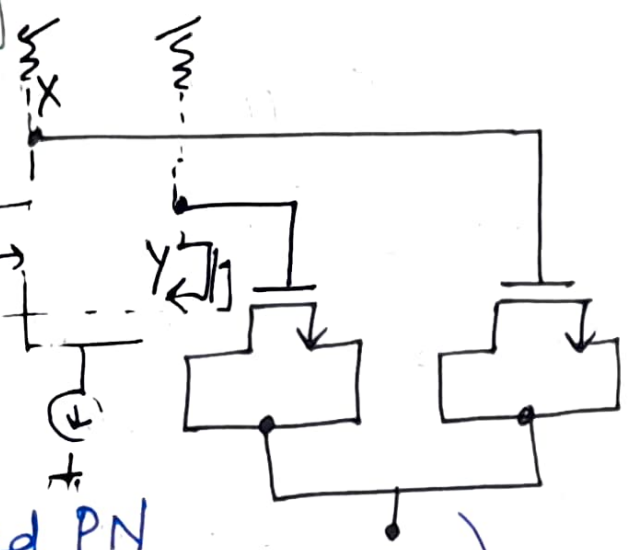
Advantages

① Negligible Added PN

Varactors introduce very little phase noise through their cont. Volt.
(MS in resistive tuning $\rightarrow$ flicker noise)

② Preferred technique

$\checkmark$ for tuning VCOs

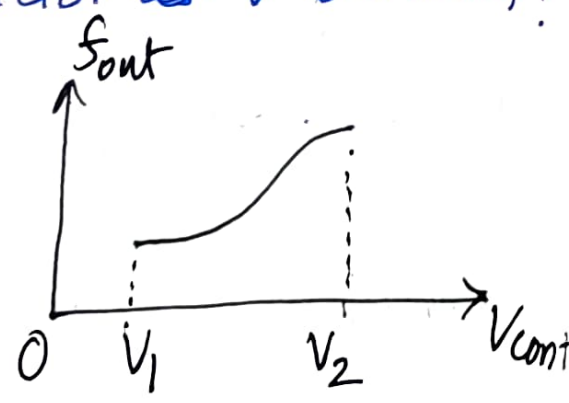
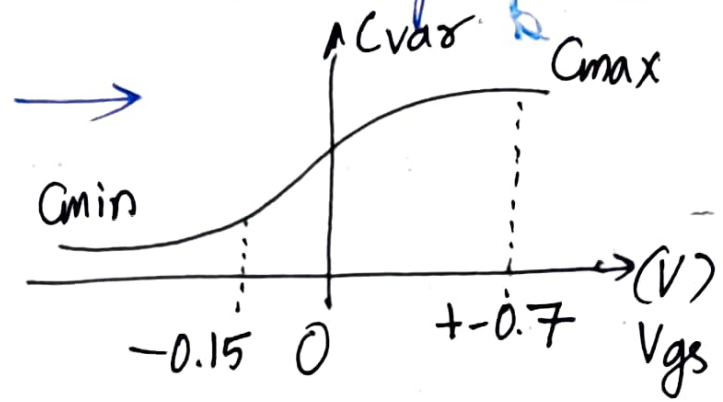


Example 4.13

if $V_{cont} \leftarrow$ varies from

$V_1 (\approx 0)$ to $V_2 (\approx V_{DD})$

which part of varactor $C-V$ is used?



* V_{gs-var} is approx common mode voltage at X/Y

$$V_{DD} - \frac{I_{SS} R_D}{2} - V_{cont}$$

$V_{cont}: V_1 \text{ to } V_2$

* $V_{cont}: V_1(\text{low}) \text{ to } V_2(\text{high})$

* V_{gs-var} : high to low $\rightarrow$ varactor o/p.

4.7.3 No. of stages

✓ vary no. of stages continuously.

✓ vary osc freq by changing the effective no. of stages.

$\rightarrow$ Each stages has $\left\{ \begin{array}{l} \text{slow path} \\ \text{fast path} \end{array} \right\}$ adjusted by V_{cont} .

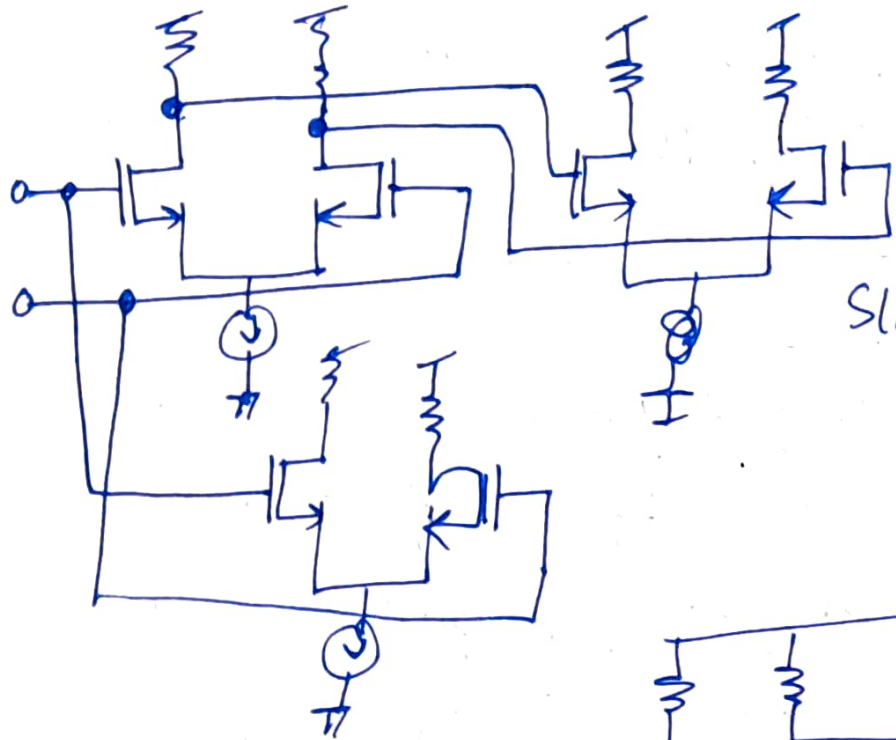
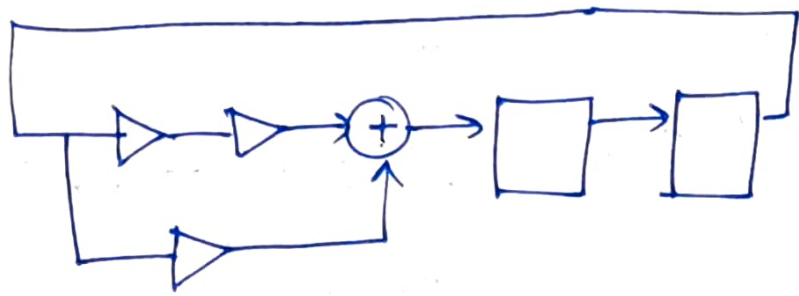
① slow path only: Fast path off $\rightarrow$ longest delay $\rightarrow$ min f_o
(V_{cont} low)

② fast path only: Fast path ON $\rightarrow$ shortest delay $\rightarrow$ max f_o
(V_{cont} high)

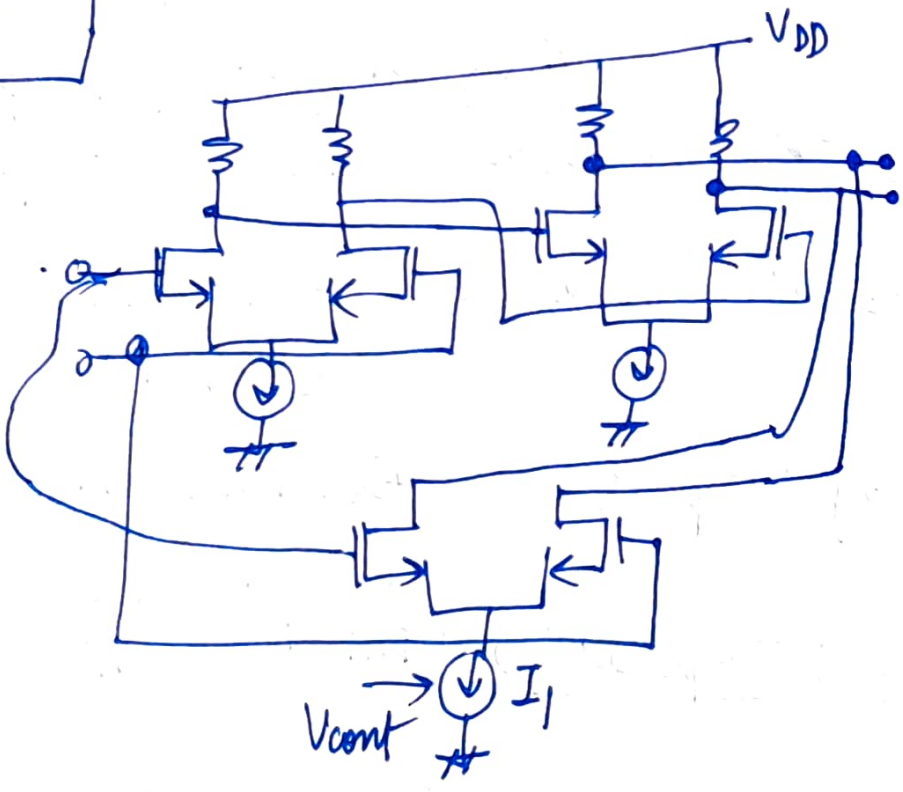
③ interpolation: Middle V_{cont} $\rightarrow$ short + long blend.

N: 6 to 3 varæes.

interpol'n
a



b
Slow & fast paths



- Drawbacks :
- ① Supply Sensitivity → current control
T enters triode
 V_{DD} affects freq.
 - ② High Phase Noise
more active pair at lower frequencies
 - ③ Tuning Device Noise noise from I_1 directly modulates freq → adds P.N.

Overall

Resistive : poor linearity/noise
Stages : wide range, high PN
Varactor : Best

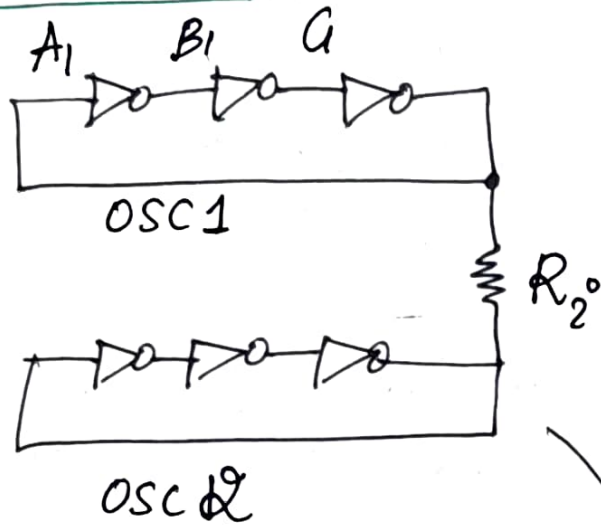
4.8 Comparison of Invt Based & Diff Based

- ① Inverter-Based → Better power & PN tradeoff
- ② diff-Based → Much less supply sensitivity.

Choose wisely

4.9 Invt Based Osc with Complementary / Quadrature outputs

4.9.1 Coupled Oscillators



Problem : Single-ended rings require odd no. of stages for oscillations
→ no direct generation of 180°

Soln : Coupled oscillators : ✓ synchronised

① Single Resistor (R_2) ←

- they tend to operate in-phase
- no current flows through R_2 . ←
- minimum ~~current~~ energy exchange.

✓ no complimentary outputs

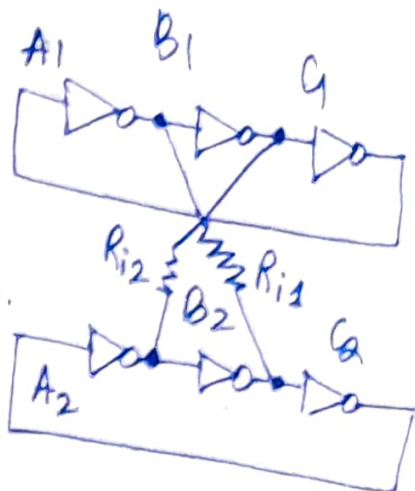
② two resistors (R_{i1}, R_{i2})

✓ in phase operation for
✓ $2/3$ of operⁿ

✓ out of phase for $1/3^{\text{rd}}$ of op.

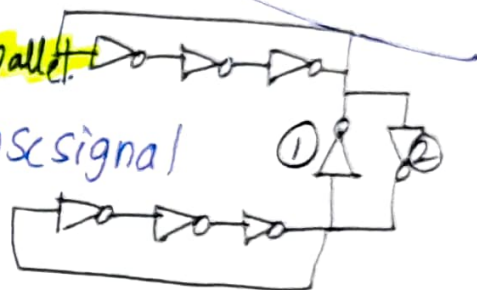
✓ circuit prefers out-of-phase
↳ less E_K exchange.

no other phase diff is stable



③ Inverter-Based

- invt & invl-2 are anti-parallel
- inject a fraction of one osc signal into another.



A_1 & A_2 out of phase by 180°

→ Invt's only inject current (transient), minimizing dissipation.

$A_1 = A_2$ Inphase is BAD

→ coupling invts have Equal in/op → fight Equal st.

In-phase



Coupling intv strength

→ Too weak : cant synchronise if
osc have freq mismatch
— o/p's become corrupted.

→ Too strong : Regenerative latch.
— latch-up.

Rule of thumb strength = 20% of main
inverters.

4.9.2 Phase Noise Considerations

• Design Goal : To design coupled oscillator
topology for 2 GHz

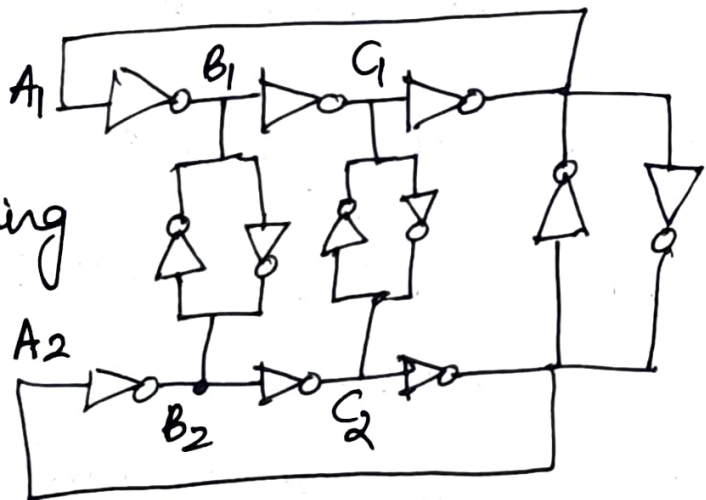
* uses stages
derived from
2.5 GHz 3-stage ring
in Chapter 3

* Simulation Results

→ $f_0 = 2.3 \text{ GHz}$

($V_{DD} = 0.95 \text{ V}$, $T = 75^\circ \text{C}$, SS corner)

* This coupled design → has 30 "unit" inverters
[6 main + 4x strength + 6 coupling invt x 1 stage]



$$= 24 + 6 = 30 \text{ unit.}$$

* Expected power/Noise tradeoff:

10 times more "unit" inverters (10x power)
 $10 \log_{10}(10) = 10 \text{ dB}$ red. in phase noise.

Actual obsv.

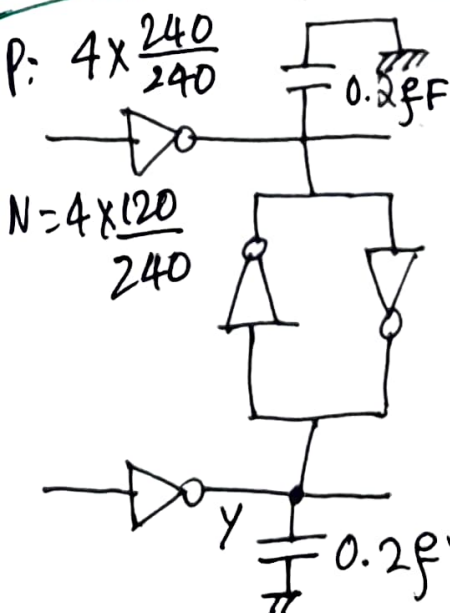
* $I = 115 \mu\text{A}$ ← supply current

* freq is lower (2.3 GHz vs 2.5 GHz)

* Result: 10 dB reduction in PN compared to simple 3-stage ring

✓ this generates complementary outputs with little penalty to FOM in terms of phase noise relative to power consumption.

Example 4.16 Does Diff. Opⁿ reduce Supply sensitivity



$$N_{\text{MOS}} = \frac{120\text{m}}{24\text{nm}}$$

$$P_{\text{MOS}} = \frac{240\text{nm}}{240\text{nm}}$$

→ does it make osc less sensitive to supply variations? of (two coupled invt rings)

• in true diff oscillators, supply noise affects both sides & gets cancelled off even when ops look complementary
 • in here, each inverter still depends on V_{DD} to switch.

• So if V_{DD} changes, all invt delays change.

→ affects freq of op.

→ shifts freq for both rings in the same direction.

→ No cancellation of that effect.

* Benefit → is small $\sim 15\%$ than simple rings

4.9.3 Dinkell Quadrature Generator (4-stage)

Goal: get A, B, C, D waveform

• $A \leftrightarrow B = \text{complementary}$

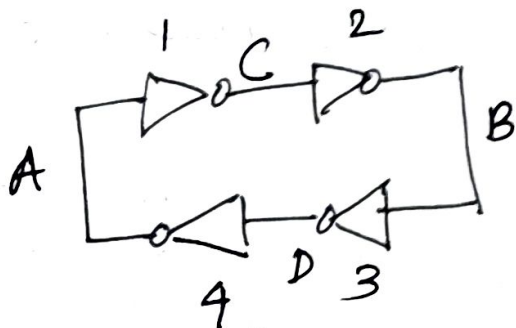
• $C \leftrightarrow D \rightarrow$

$A \& C = 90^\circ \text{ apart (quadrature)}$

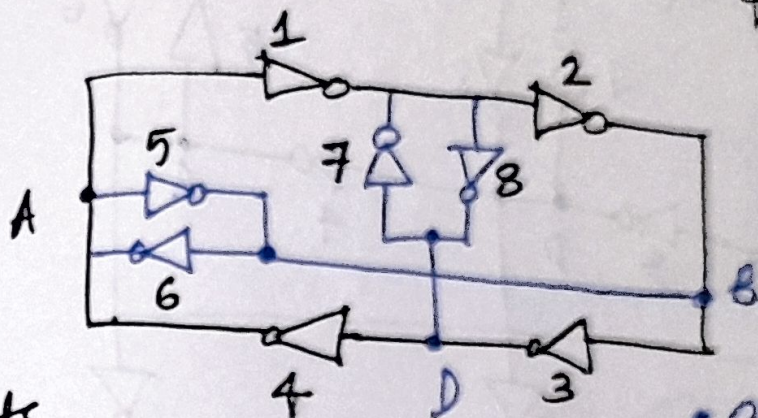
✓ uses 4 stages in a ring.

✓ each invt adds delay → 90° apart .

✓ Problem: Latchup possible.



Solution is to add cross-coupled inverters
 int5 - invt8 to prevent latch up.
 → fight equal states & force transitions.



Keep 5 to 8 inv.
 weaker than
 main inverters

Rule of Thumb:

Results

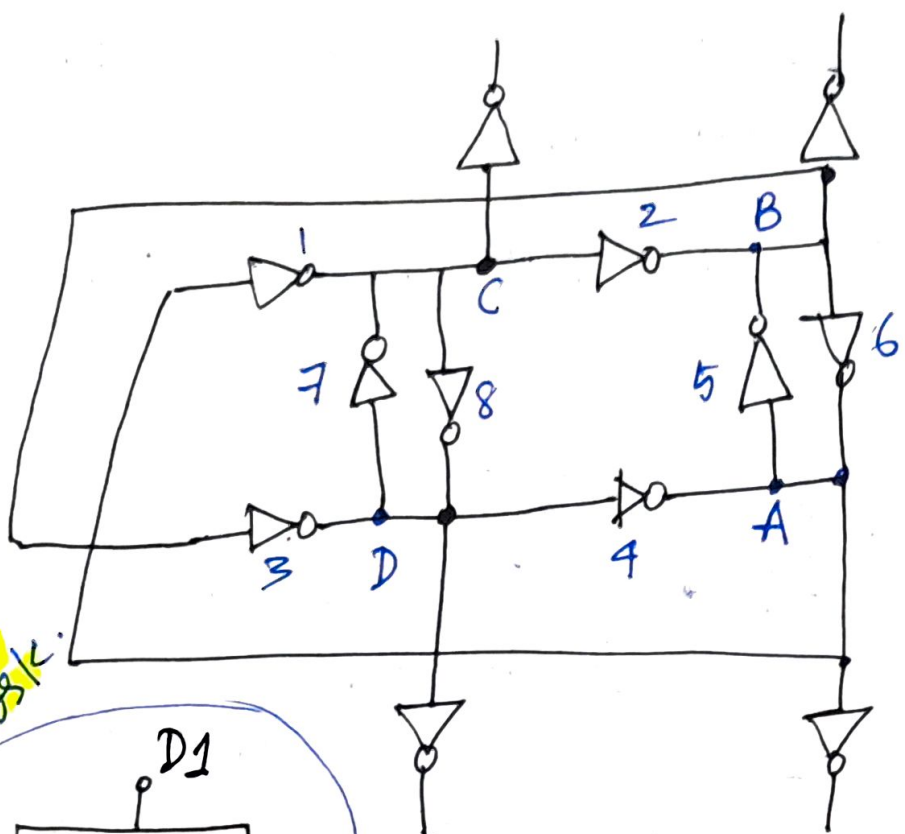
- Higher freq than 3-stg keep it 2:1
- Slightly worse FOM due to extra power & noise
- gives True quadrature o/p's

Why?

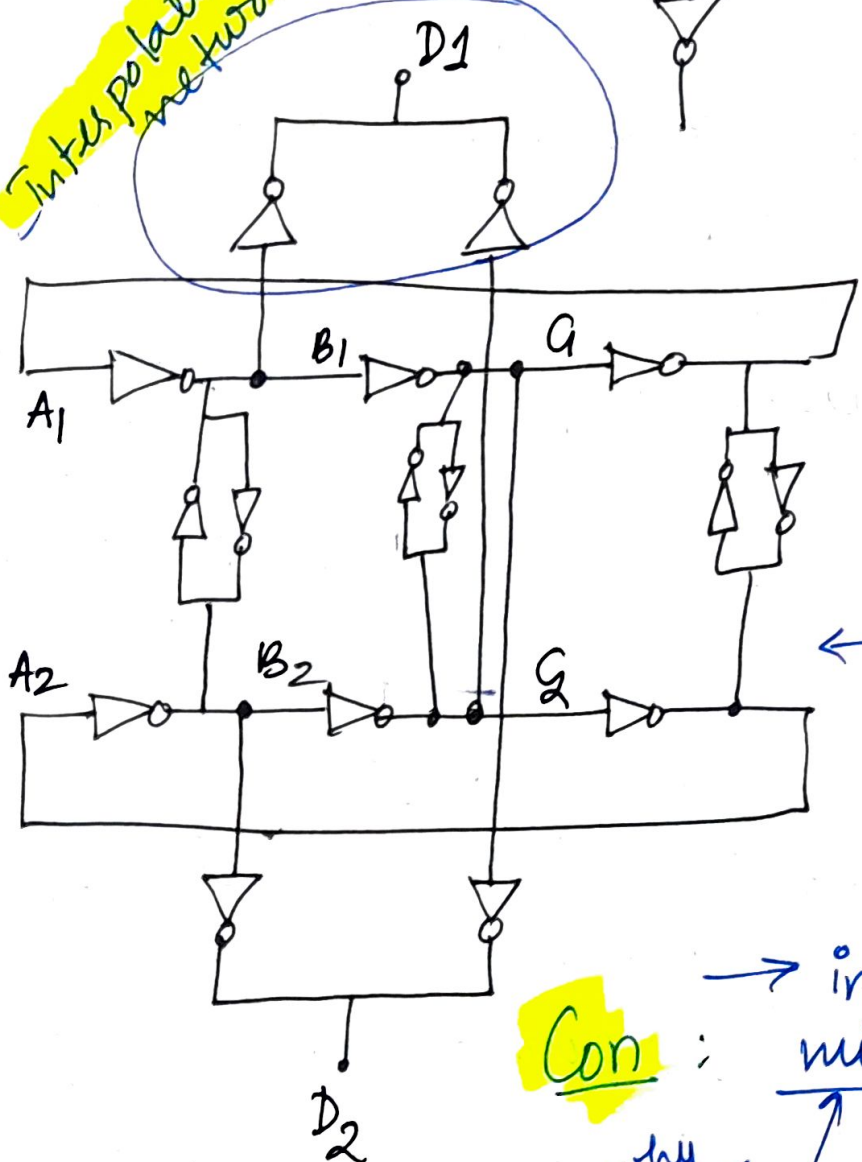
- too weak: can't correct mismatch or prevent latch
- too strong: will overpower & cause latchup

4.9.4 Quadrature by interpolation

- goal is to create immediate phases (like 90° shifts) without special topology.
- Working: take output from two inverters (say B1 & C3)
 - : feed them into 2-inverters & short their o/p's
 - : the result: $\frac{B2 + C2}{2}$; a signal halfway b/w quadrature!



Interpolation
not two 8k.



only draws static current when $B1 \neq C2$
 → brief fight helps interpolation
 ← Inverters don't always fight only when signal to exist.

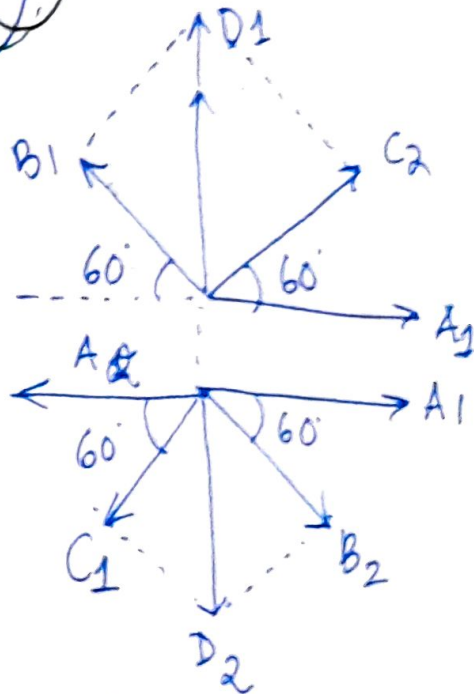
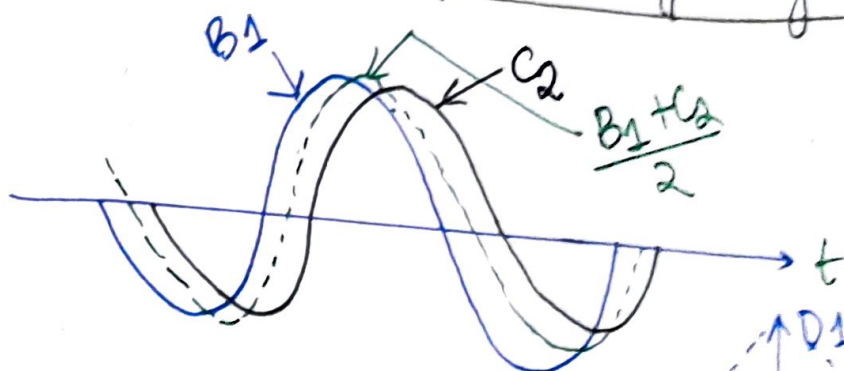
Con: introduces Load mismatch & delays

why by interpolation of int 1 & 2
 in inverters b/w leads to phase
 mismatch b/w
 $B1 \& A1$ & $B2 \& A2$
 $D1, A1$
 $D2, A2$

how to fix?

D_1 to D_4 : 4 o/p's serve as quadrature phases

introduces phase mismatches.
Be careful in layout & symm.



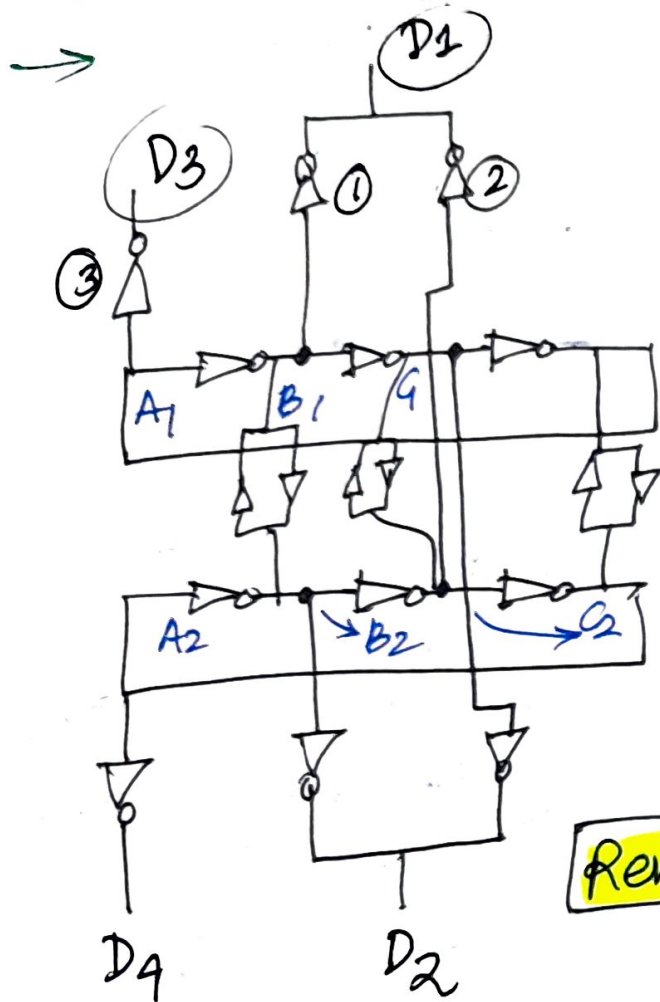
Summary

Method	Acce.	Complex	Power	
Direct 4-st	high	Moderate	higher	Robust, needs cross-coupling
interpolation	Moderate	Simple	Lower	fine tuning needed less area.

Example 4.17
Suppose D_1 & D_3 drive Equal load cap.
Do these signals experience phase mismatch?

Example 4.1
Suppose D_1 & D_3 drive Equal load Cap.
excessive phase mism

Suppose V_1 & V_3 are out of phase.
Do these signals experience phase mismatch?



→ driving strength is greater at D_1 than D_3 .

→ D_1 is formed by 2 investors.

Remedy : fanout seen at these o/p's should be small.

i.e., int 1 to 3 are
several times larger
than the subsequent stg

no need of larger Transu

NOTE

negligible degradation
of phase noise

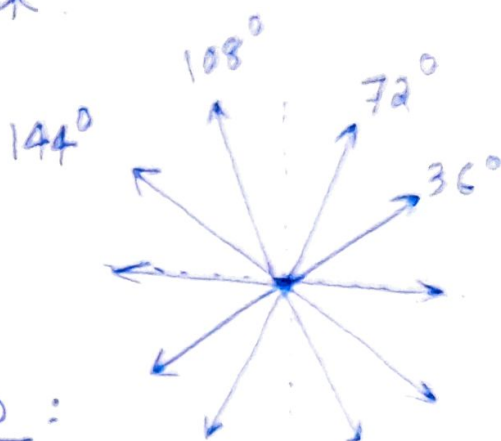
- they (interpolⁿ inpts) stay out of osc loops, there's P.N. is not shaped by $(b/f)^2 \rightarrow$ Remains small.

Remains
Small. 4

Example 4.18

2: 5-stage ring os are coupled so as to operate 180° out of phase. How shld quadrature wave-forms be generated?
→ 4 & quick

- $36^\circ \leftarrow \frac{360^\circ}{10}$
↑ phase resolution of 2-rings.



- For 90° resolution:

we can interpolate b/w 36° & 144° o/p.s
or 72° & 108° o/p.s

preferred

why???

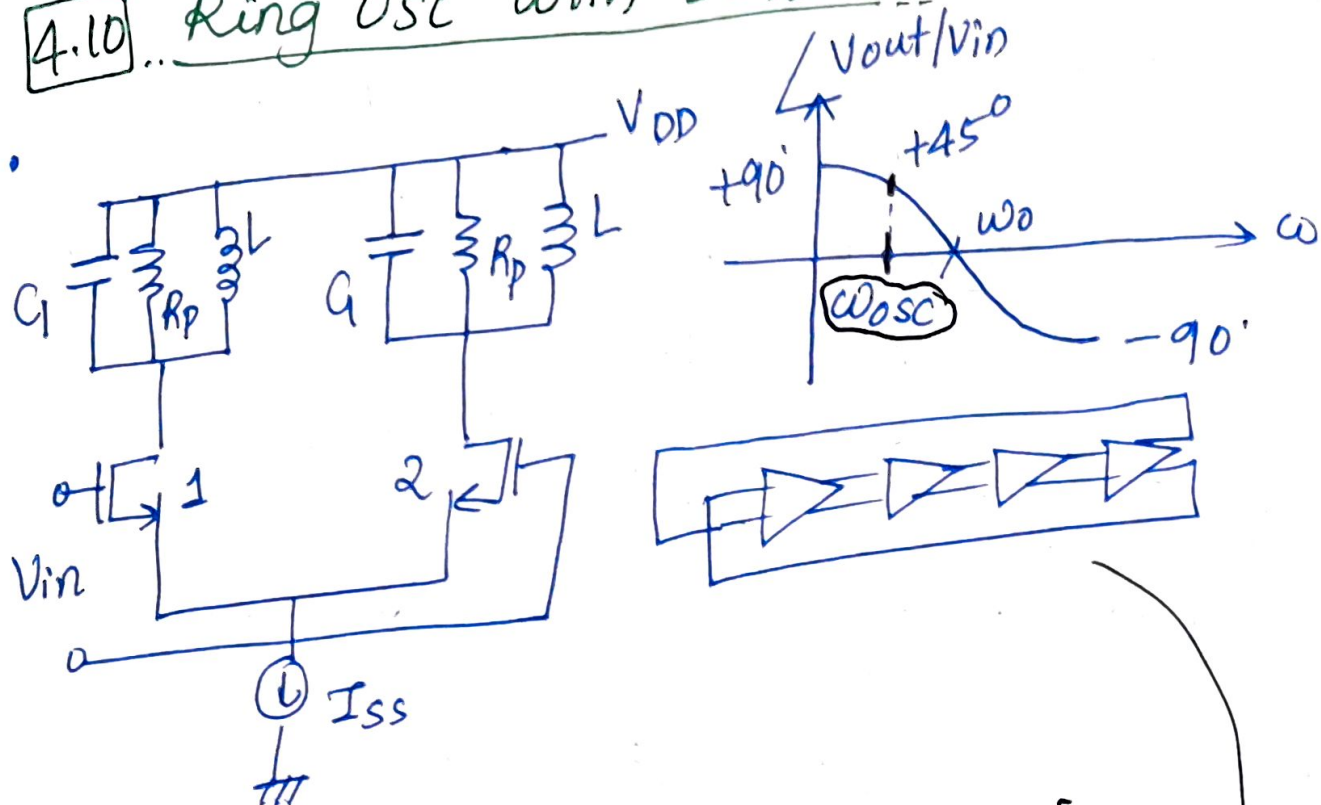
→ smaller phase diff

interpolating

inverters fight

to a lesser extent

4.10 Ring OSC with LC loads



- feedback is -ve at low frequencies.
 - Barkhausen's criterion dictates the 4-stage
 - 180° total phase shift at osc freq (ω_{osc})
 - resonance freq exhibited by each stage
- $\hookrightarrow \omega_0 = \frac{1}{\sqrt{L_1 C_1}}$ but no phase shift at ω_0

✓ osc freq must depart from ω_0 so as to yield a phase shift of 45° /stage.

To Calculate ω_{osc} , equal phase shift of parallel RLC = 45° .

$$\frac{\pi}{2} - \tan^{-1} \left(\frac{L\omega_{osc}}{R_p(1 - L_1 C_1 \omega_{osc}^2)} \right) = \frac{\pi}{4}$$

approx $\frac{R_p}{L\omega_{osc}}$ by Q at ω_0

$$\therefore \omega_{osc} = \sqrt{1 - \frac{1}{Q^2 \sqrt{L_1 C_1}}}$$

✓ if $\Delta\omega = \omega_0 - \omega_{osc}$ is small.

$$Q = \left(\frac{\omega_0}{2} \right) \frac{d\phi}{d\omega} \quad d\phi \approx \frac{\pi}{4}$$

$$\therefore \Delta\omega \approx \frac{\pi}{8Q} \omega_0$$

Advantages

- ① Low supply Operation: $\rightarrow$ inductive loads drop little DC voltage. $\rightarrow$ i/o CM levels stay near V_{DD} , $\rightarrow$ low voltage operation.
- ② Frequency stability with stage Count
 $\rightarrow$ osc freq depends weakly on N-stages. $\Delta\omega = \frac{\pi \omega_0}{2QN}$
 $\rightarrow$ N increases $\rightarrow$ freq approaches ω_0
- ③ higher frequency:
 $\rightarrow$ load cap is absorbed into LC tanks.
 $\rightarrow$ allows higher freq osc than tradⁿ RC-load rings

Drawback ✓ requires multiple spiral inductors increasing area & design complexity.
 $\rightarrow$ can be partly mitigated using stacked inductor structures.